

Clase: Regresión Parte I

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Modelo de Regresión Simple

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¿Qué es Econometría?

- *Utilizar métodos de regresión múltiple para contrastar teorías con datos empíricos, generar predicciones.*
- *Aplicaciones en economía urbana: Evaluación de impacto (programas de titulación, infraestructura, vivienda, crimen, transporte), Modelo de valuación de propiedades, establecer las tasas sobre la propiedad.*
- *Otras aplicaciones popular en economía: Predicciones de modelos macroeconómicos.*

Pasos en el Análisis Empírico

- *Establecer con cuidado la pregunta.*
- *Modelo teórico (modelo marco de maximización de la utilidad, beneficios de la firma etc)*
- *Cuál es la variable a explicar? (En infraestructura, efectos en salud, escolaridad. En modelos de precios hedónicos, valores de venta y alquiler)*

Pasos en el Análisis Empírico

- *1- Especificación del modelo económico*
- *Ej 1: (Modelo de participacion en el crimen a la Becker)*

$$y = f(x_1, x_2, x_3, x_4, x_5, x_6, x_7)$$

y _ horas dedicadas a actividades criminales

x_1 _ Salario por una hora de actividad criminal

x_2 _ salario por una hora dedicada al empleo

x_3 _ otros ingresos

x_4 _ Probabilidad de ser atrapado

x_5 _ Probabilidad de ser encarcelado

x_6 _ Sentencia esperada

x_7 _ edad

Pasos en el Análisis Empírico

- *1- Especificación del modelo económico*
- *Ej 2: (Modelo de Precios Hedónicos, versiones similares en Cruces et al, Olmos, Gulyani et al)*

Precioinmueble

$$= f(\underbrace{\text{superficie}}_{+}, \underbrace{\text{tamaño del lote}}_{+}, \underbrace{\text{dormitorios}}_{+}, \underbrace{\text{edad}}_{-}, \underbrace{\text{gar}}_{-})$$

Pasos en el Análisis Empírico

- *2- Definición del modelo econométrico*

$$\text{precio}_i = \beta_0 + \beta_1 * \text{superficiem2}_i + \beta_2 * \text{lotemts}_i + \beta_3 * \text{ndorms}_i \\ + \beta_4 * \text{edadanos}_i + \beta_5 * \text{dgarage}_i + \beta_6 * \text{distancia}_i + \varepsilon_i$$

- $\beta_0, \beta_1, \dots, \beta_6$ *parámetros*
- ε_i *término de error*
- *3- Definición de hipótesis*

Estructura de una base de datos

- 1. Corte Transversal*
- 2. Series de Tiempo*
- 3. Pool y Datos de Panel*

A Cross-Sectional Data Set on Wages and Other Individual Characteristics

<i>obsno</i>	<i>wage</i>	<i>educ</i>	<i>exper</i>	<i>female</i>	<i>married</i>
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

Minimum Wage, Unemployment, and Related Data for Puerto Rico

<i>obsno</i>	<i>year</i>	<i>avgmin</i>	<i>avgcov</i>	<i>unemp</i>	<i>gnp</i>
1	1950	0.20	20.1	15.4	878.7
2	1951	0.21	20.7	16.0	925.0
3	1952	0.23	22.6	14.8	1015.9
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
37	1986	3.35	58.1	18.9	4281.6
38	1987	3.35	58.2	16.8	4496.7

Pooled Cross Sections: Two Years of Housing Prices

<i>obsno</i>	<i>year</i>	<i>hprice</i>	<i>proptax</i>	<i>sqrft</i>	<i>bdrms</i>	<i>bthrms</i>
1	1993	85500	42	1600	3	2.0
2	1993	67300	36	1440	3	2.5
3	1993	134000	38	2000	4	2.5
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
250	1993	243600	41	2600	4	3.0
251	1995	65000	16	1250	2	1.0
252	1995	182400	20	2200	4	2.0
253	1995	97500	15	1540	3	2.0
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
520	1995	57200	16	1100	2	1.5

No necesariamente tienen que repetirse las observaciones

A Two-Year Panel Data Set on City Crime Statistics

<i>obsno</i>	<i>city</i>	<i>year</i>	<i>murders</i>	<i>population</i>	<i>unem</i>	<i>police</i>
1	1	1986	5	350000	8.7	440
2	1	1990	8	359200	7.2	471
3	2	1986	2	64300	5.4	75
4	2	1990	1	65100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260700	9.6	286
298	149	1990	6	245000	9.8	334
299	150	1986	25	543000	4.3	520
300	150	1990	32	546200	5.2	493

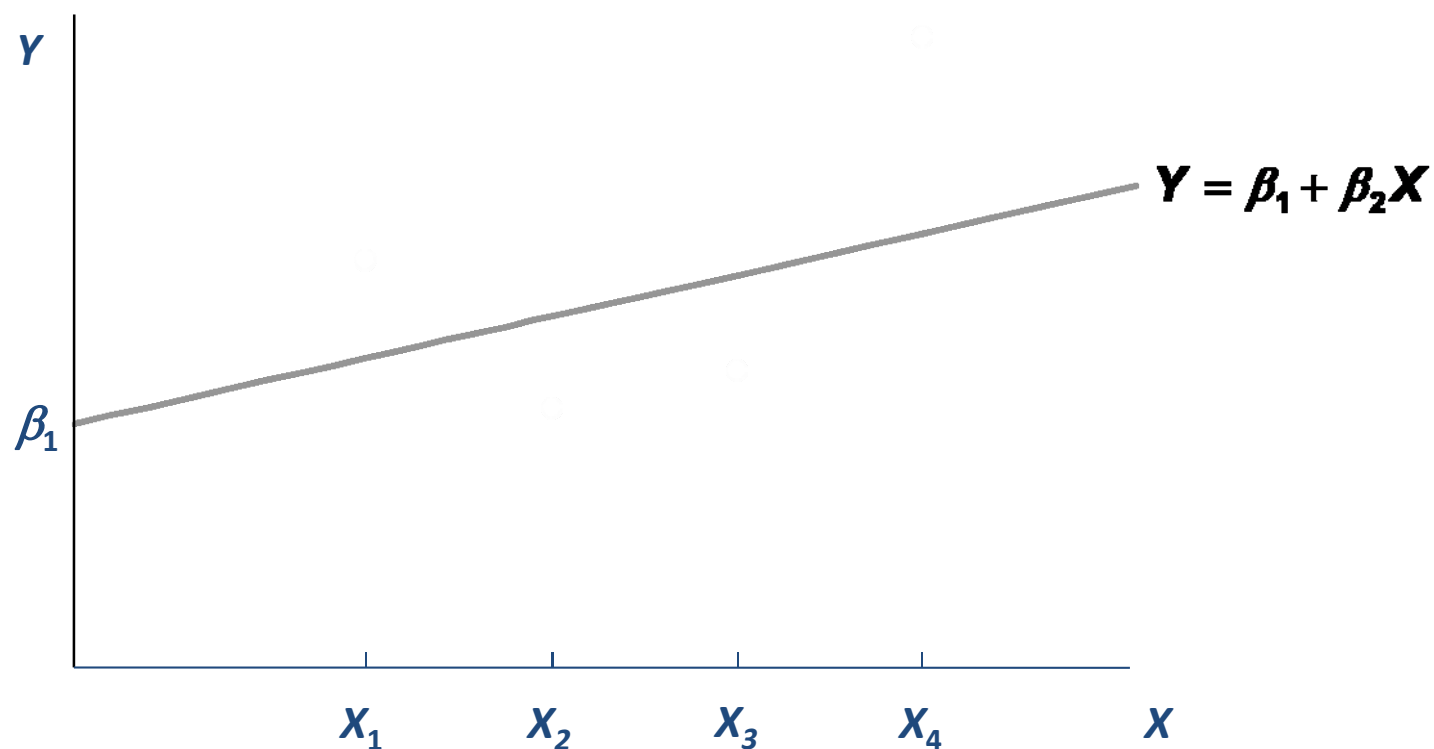
Causalidad (Brevísima Introducción)

- *El objetivo de la mayor parte de los estudios es causalidad. Se buscá más que medir correlación.*
- *El análisis **Ceteris Paribus** es crucial. (x ejem: efecto precio sobre demanda, regulación sobre informalidad) Si otros factores cambian, no identificamos causalidad.*
- *¿Se tomaron suficiente cantidad de factores de control en consideración?*

Causalidad (Brevísima Introducción)

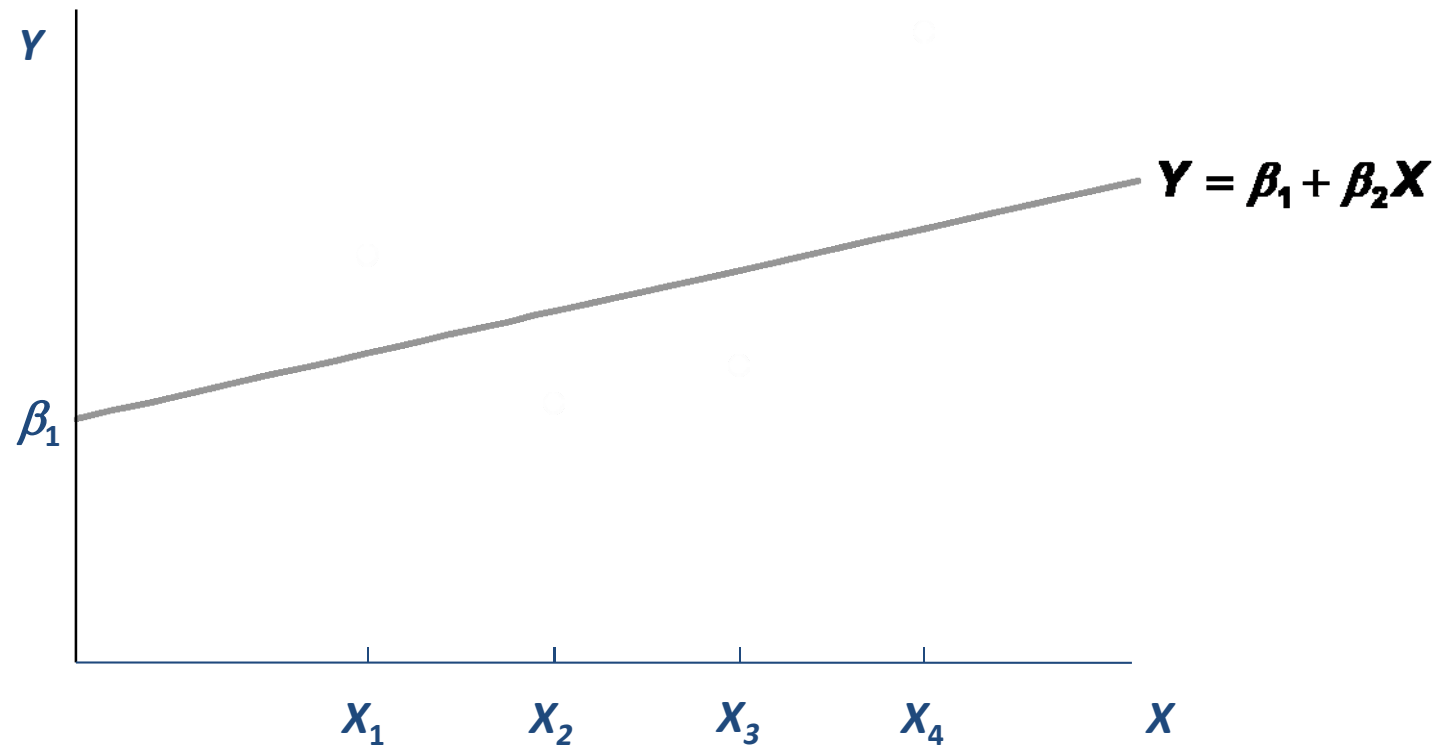
- *Ejemplo más simple: Efecto de fertilizantes en producción de soja. Como medimos?*
- *Problema: calidad de la tierra puede no ser observable*
- *Solución: Asignar aleatoriamente los lotes a fertilizar.*
- *Ejemplo 2: Medición del Efecto de un año más de Educación sobre el ingreso. Dificultades para generar un experimento. Uso de datos observacionales y sesgo de selección.*

Modelo de Regresión Simple



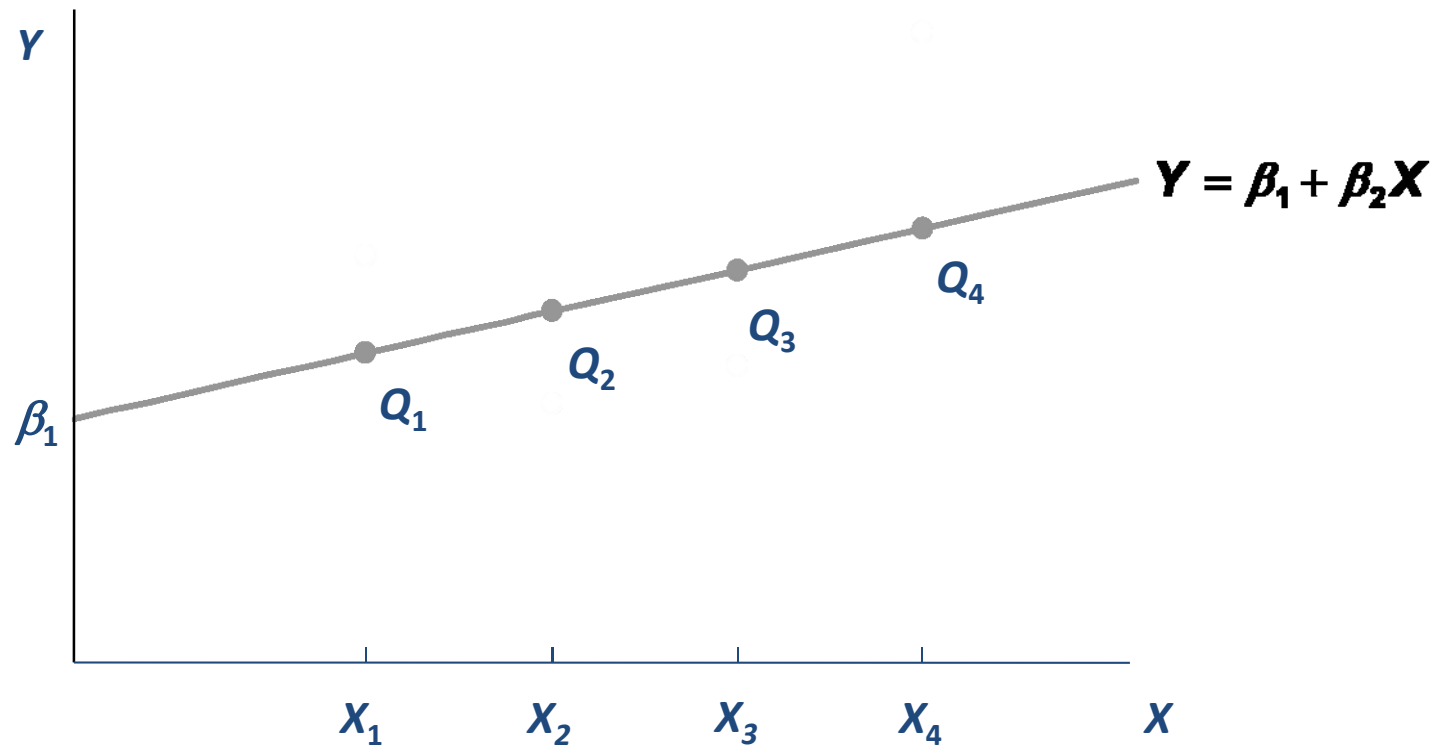
Suppose that a variable Y is a linear function of another variable X , with unknown parameters β_1 and β_2 that we wish to estimate.

Modelo de Regresión Simple



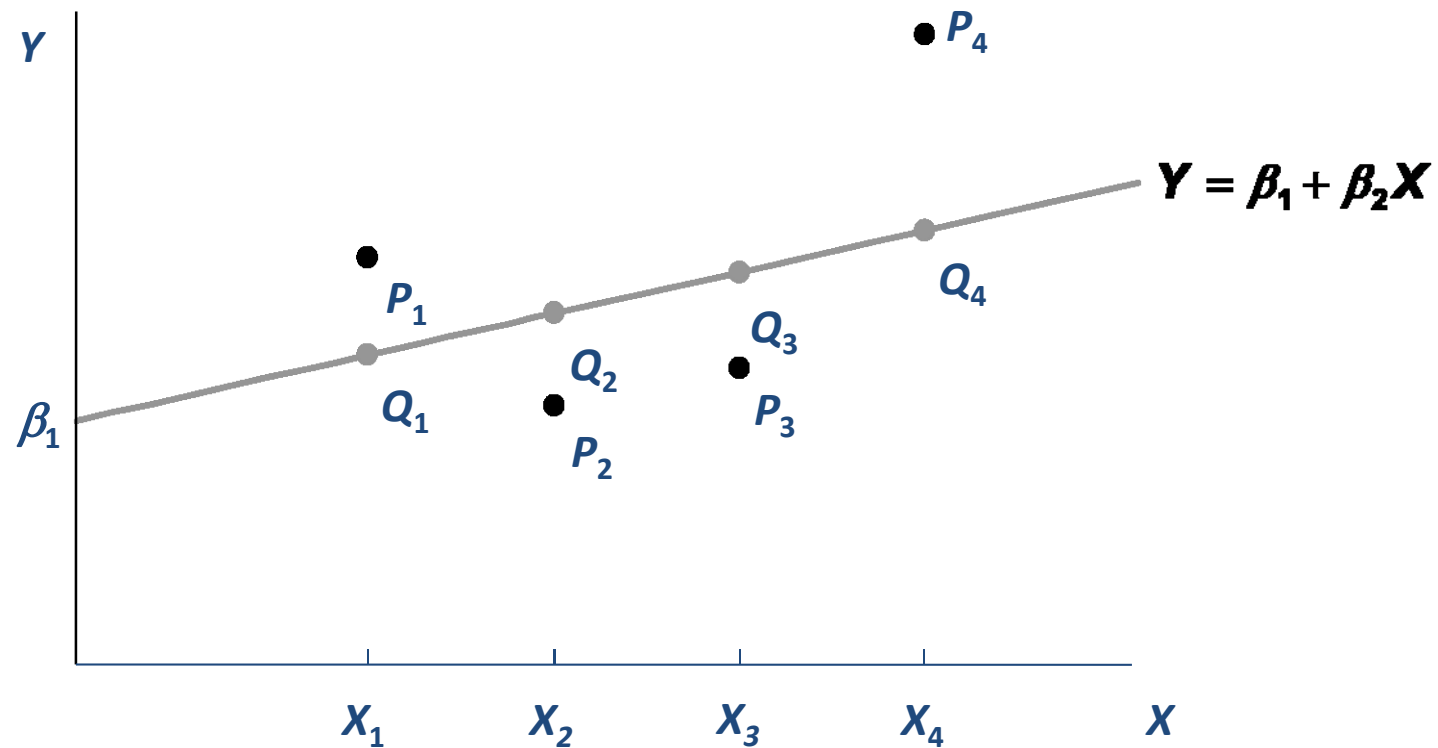
Suppose that we have a sample of 4 observations with X values as shown.

Modelo de Regresión Simple



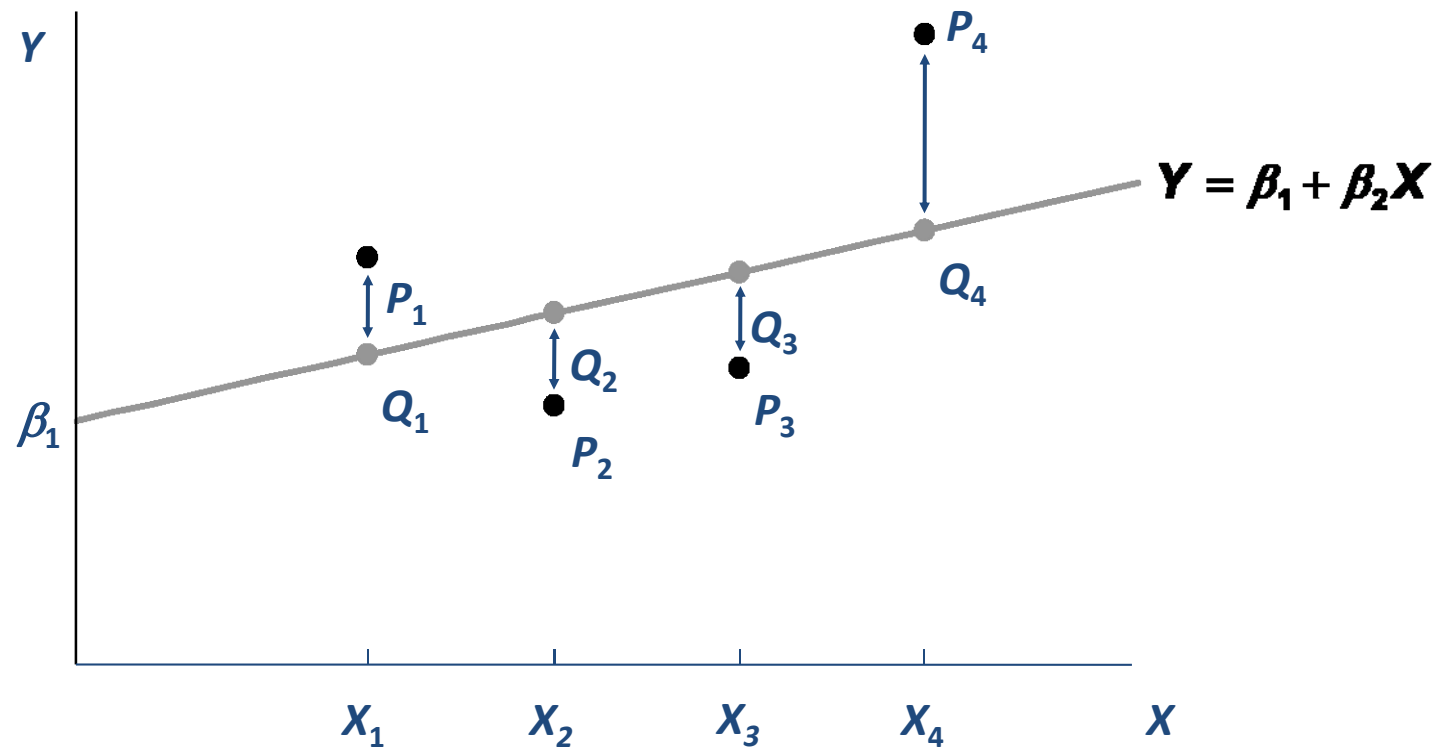
If the relationship were an exact one, the observations would lie on a straight line and we would have no trouble obtaining accurate estimates of β_1 and β_2 .

Modelo de Regresión Simple



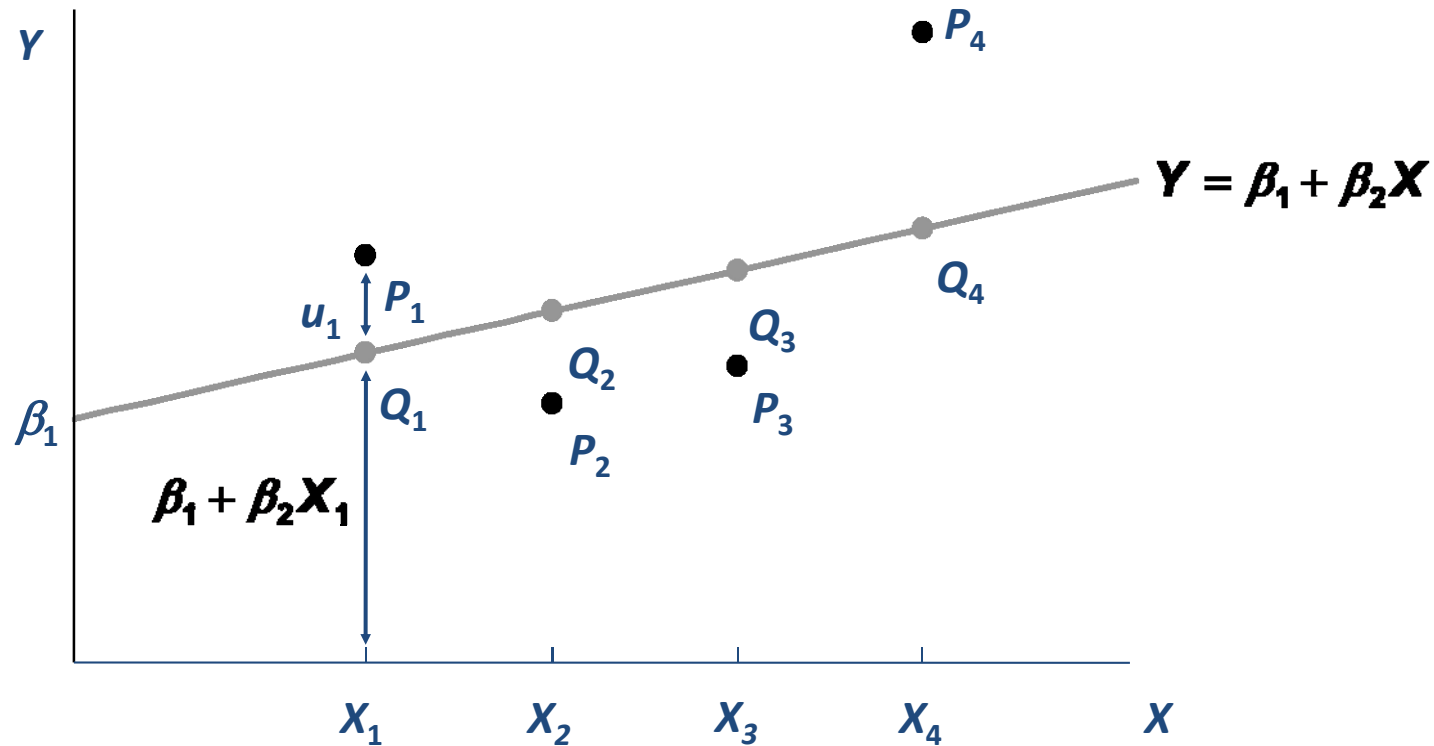
In practice, most economic relationships are not exact and the actual values of Y are different from those corresponding to the straight line.

Modelo de Regresión Simple



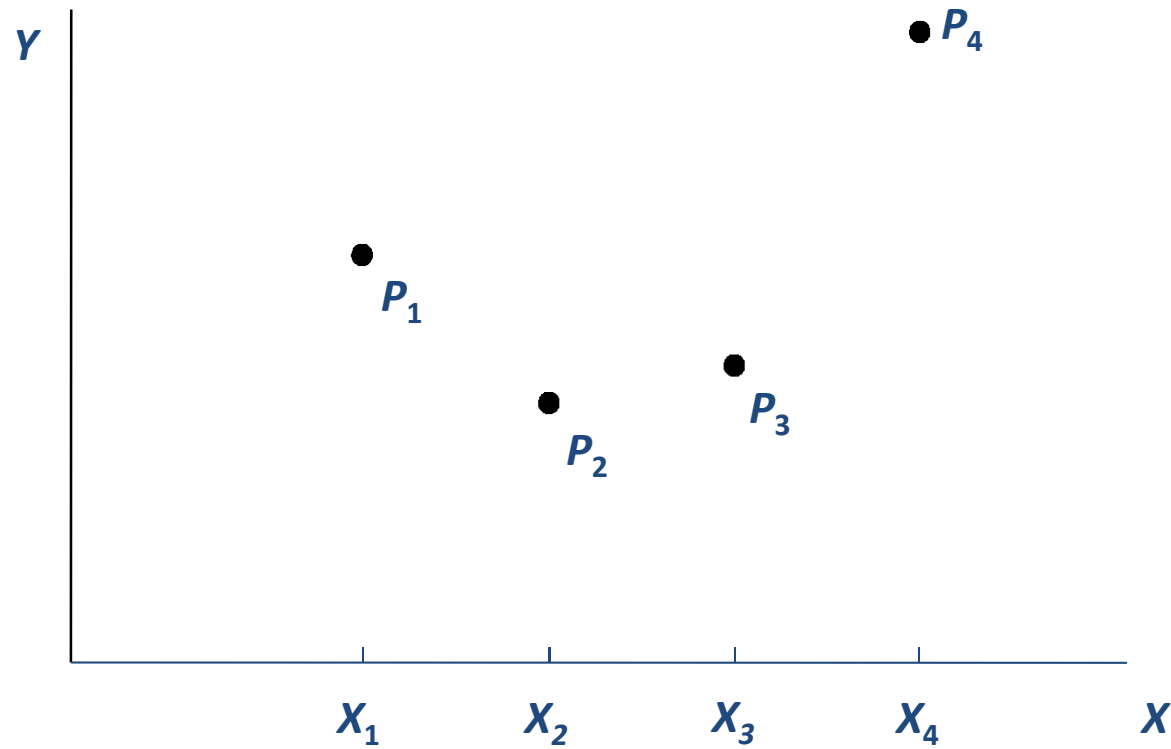
To allow for such divergences, we will write the model as $Y = \beta_1 + \beta_2 X + u$, where u is a disturbance term.

Modelo de Regresión Simple



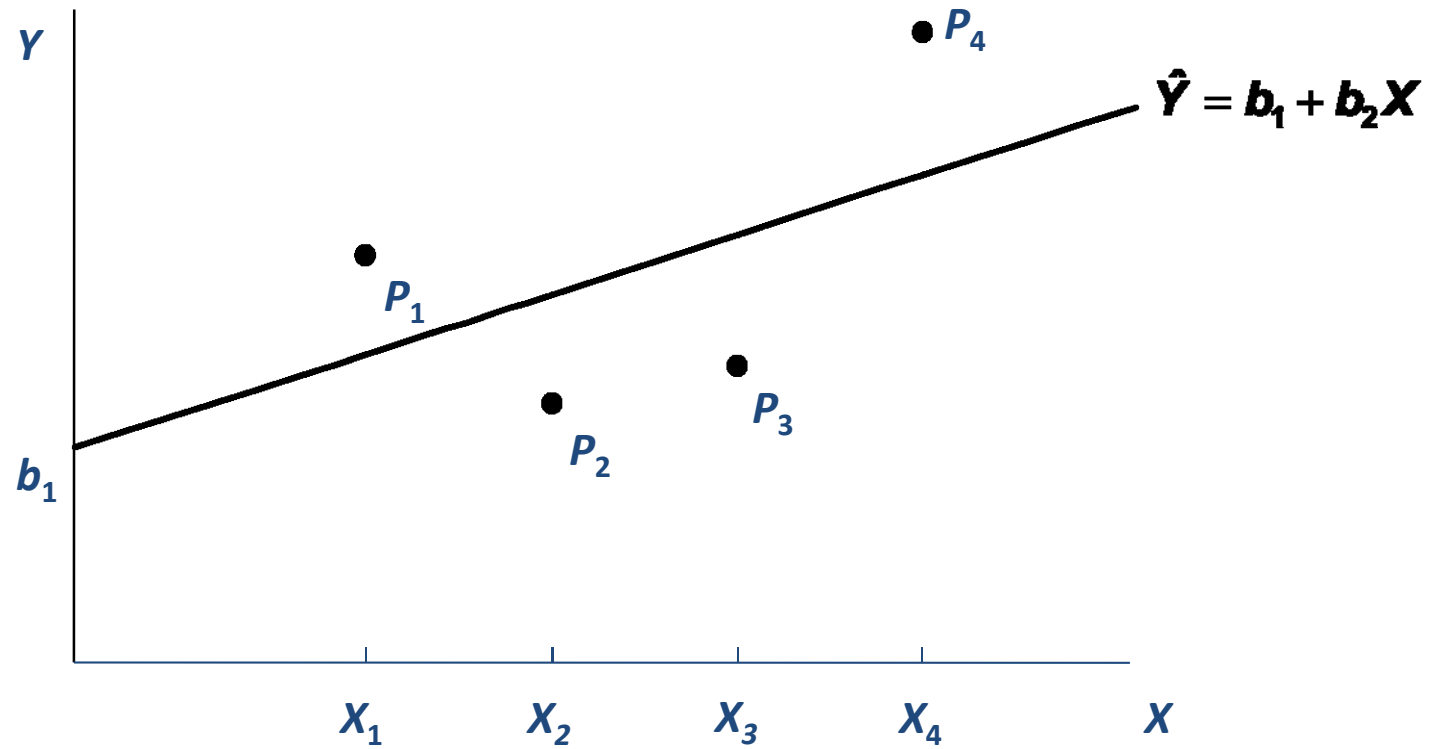
Each value of Y thus has a nonrandom component, $\beta_1 + \beta_2 X$, and a random component, u . The first observation has been decomposed into these two components

Modelo de Regresión Simple



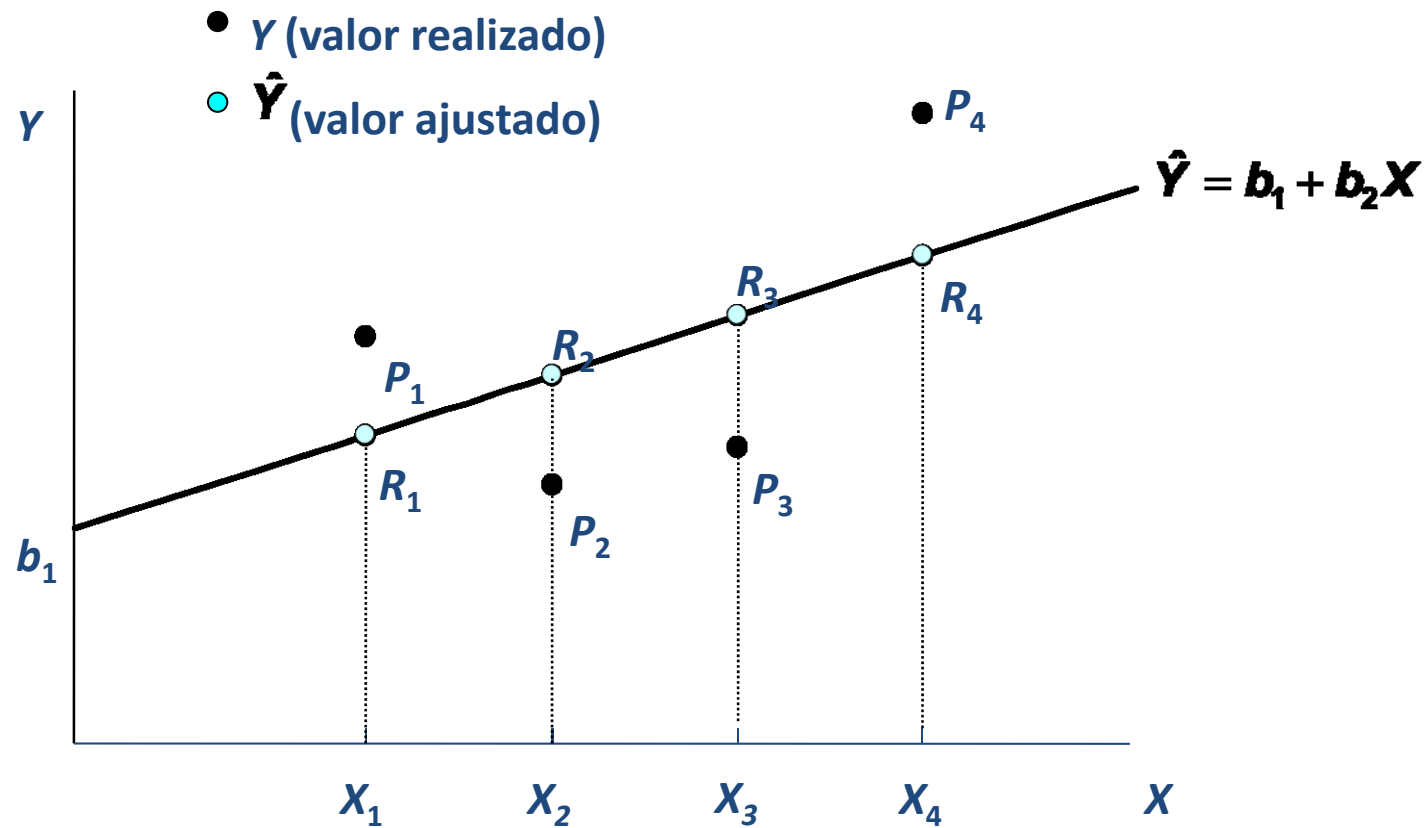
In practice we can see only the P points.

Modelo de Regresión Simple



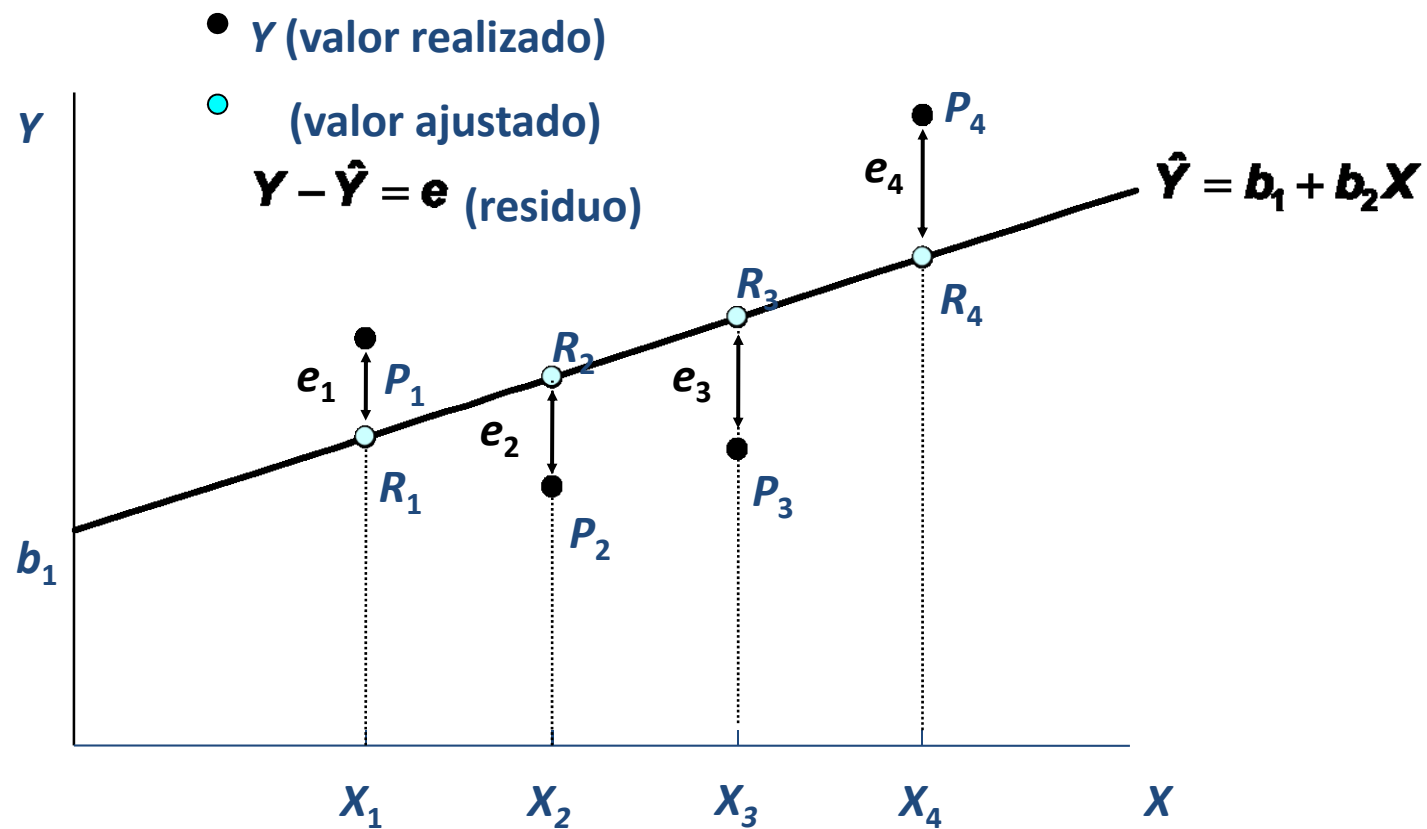
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Modelo de Regresión Simple



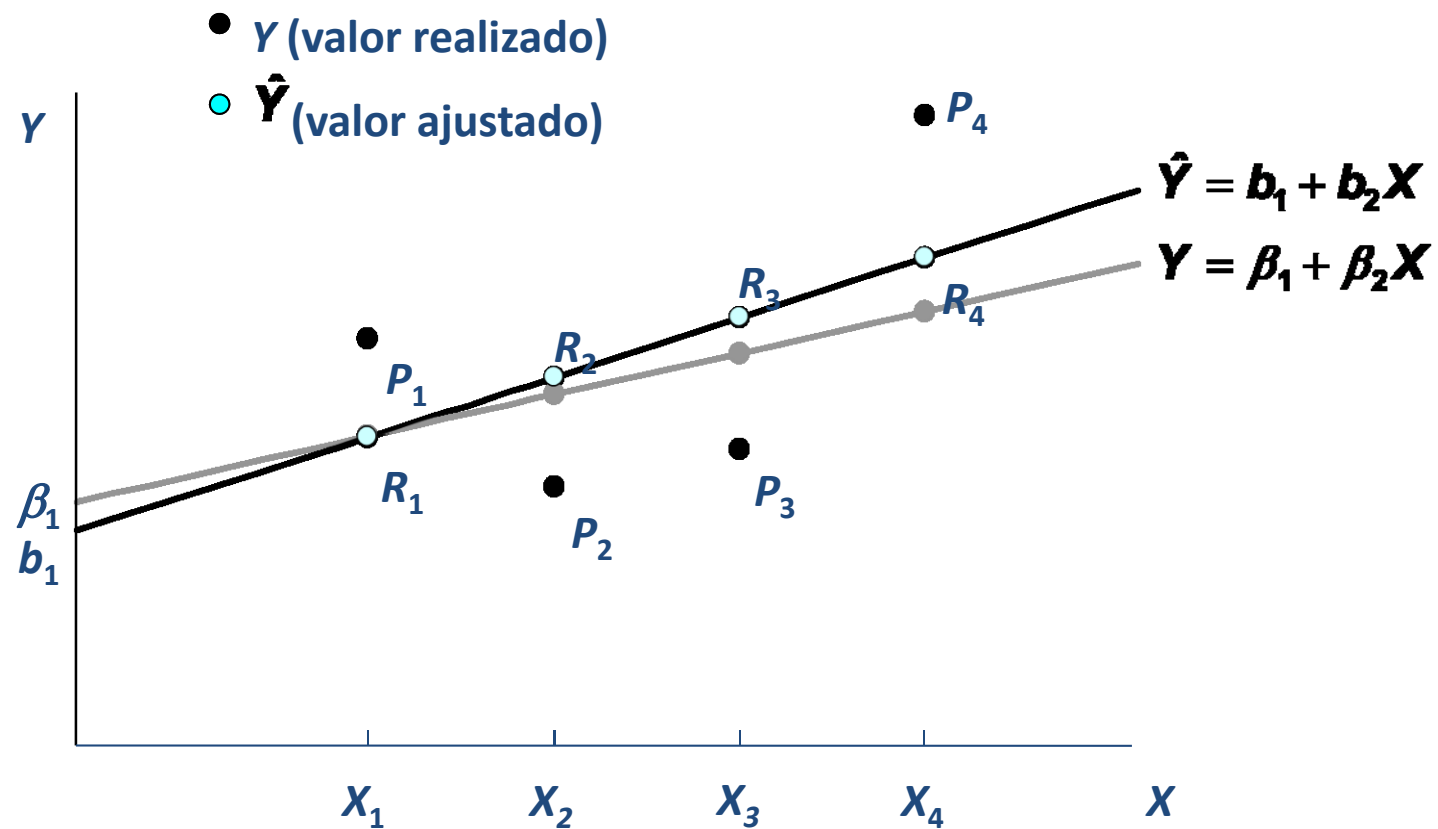
The line is called the fitted model and the values of Y predicted by it are called the fitted values of Y . They are given by the heights of the R points.

Modelo de Regresión Simple



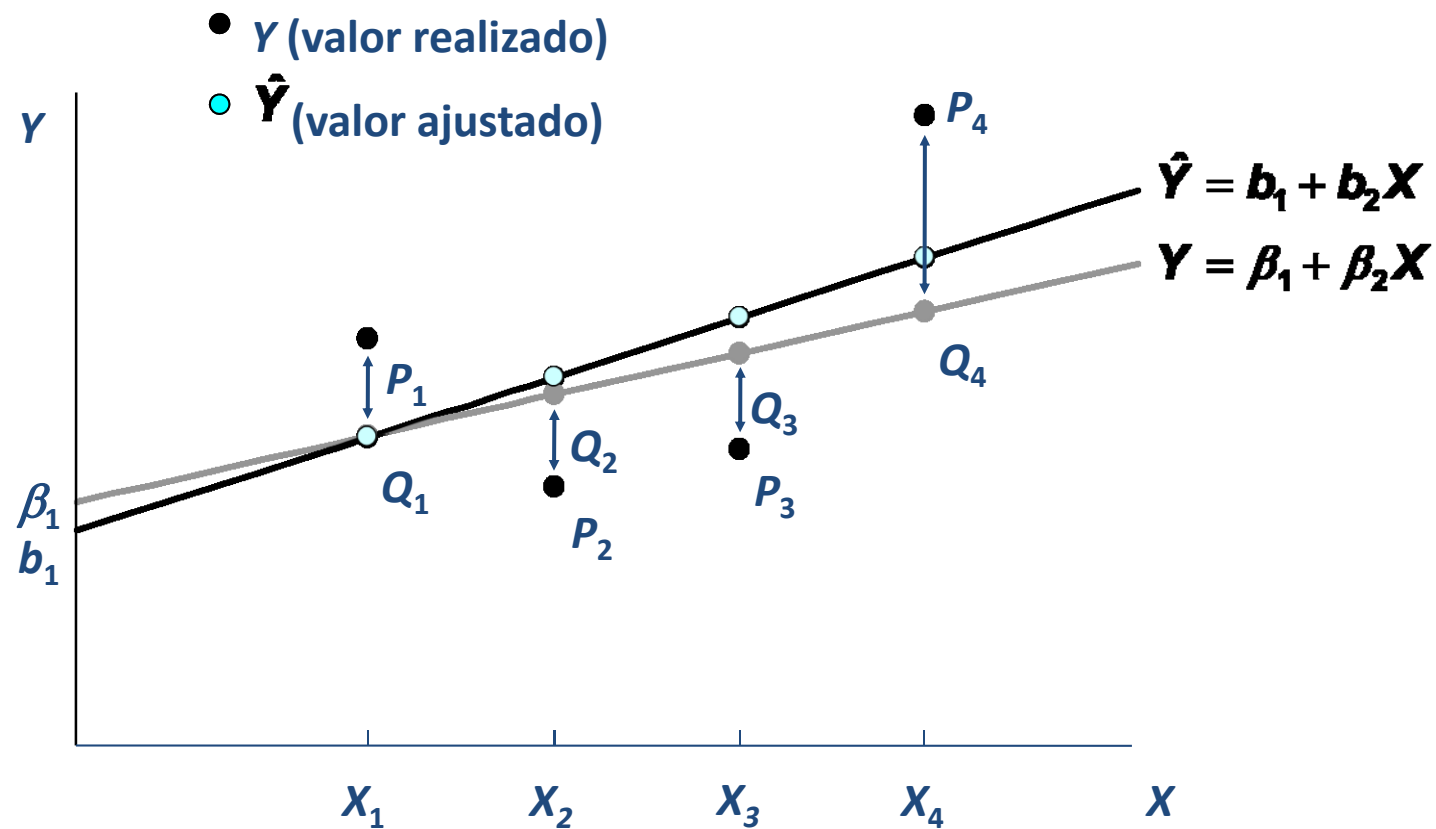
The discrepancies between the actual and fitted values of Y are known as the residuals

Modelo de Regresión Simple



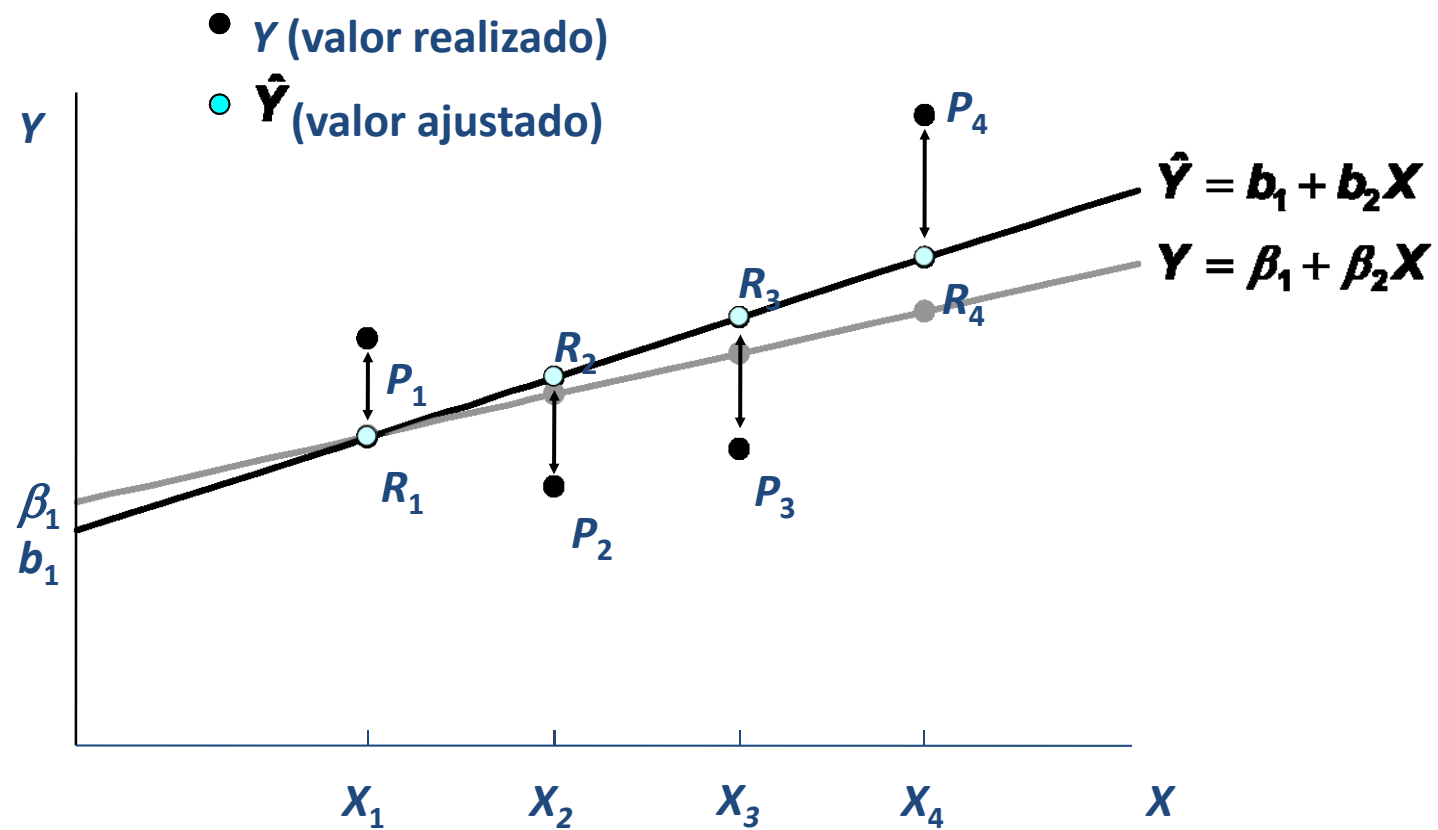
Note that the values of the residuals are not the same as the values of the disturbance term. The diagram now shows the true unknown relationship as well as the fitted line.

Modelo de Regresión Simple



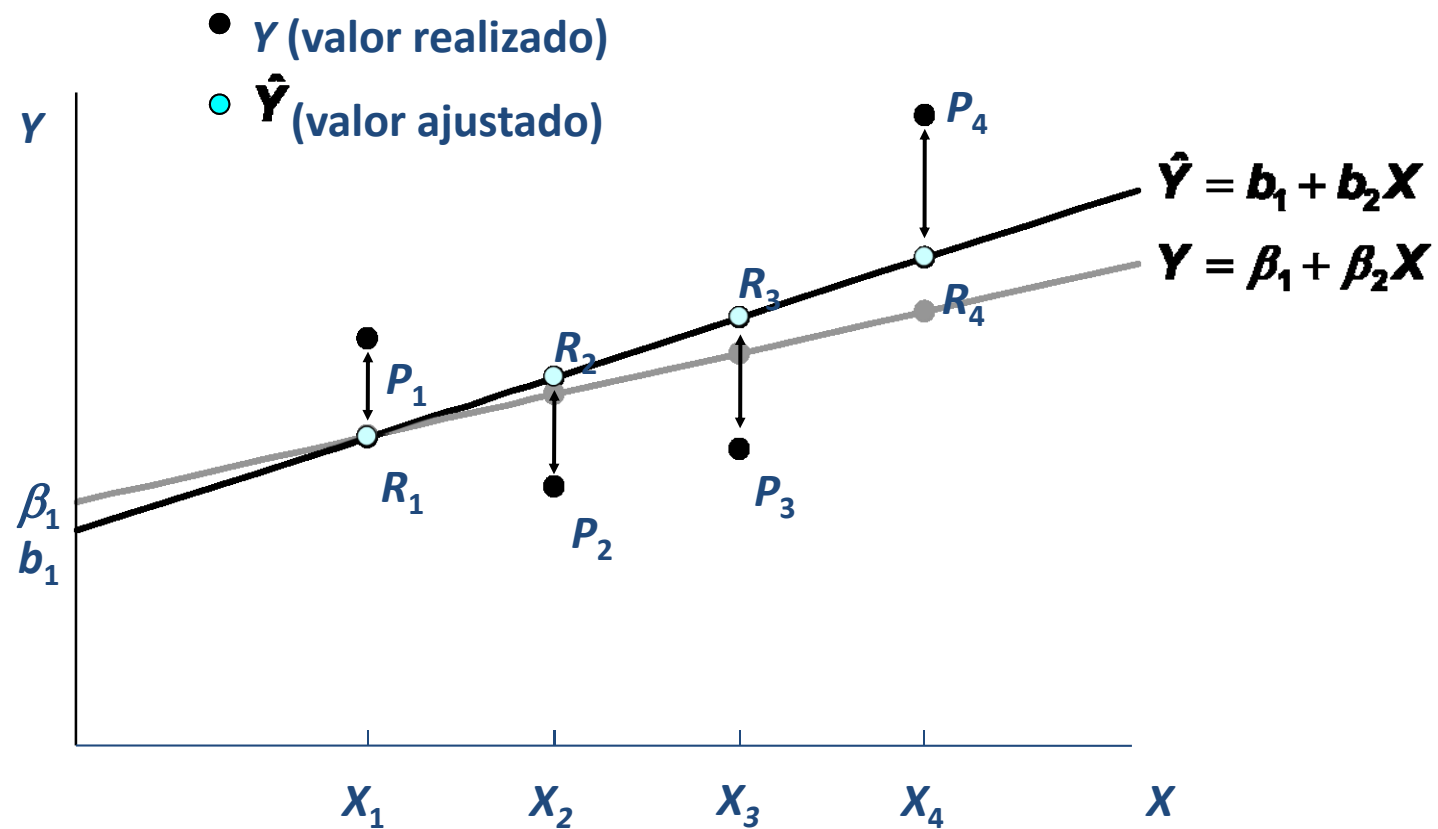
The disturbance term in each observation is responsible for the divergence between the nonrandom component of the true relationship and the actual observation.

Modelo de Regresión Simple



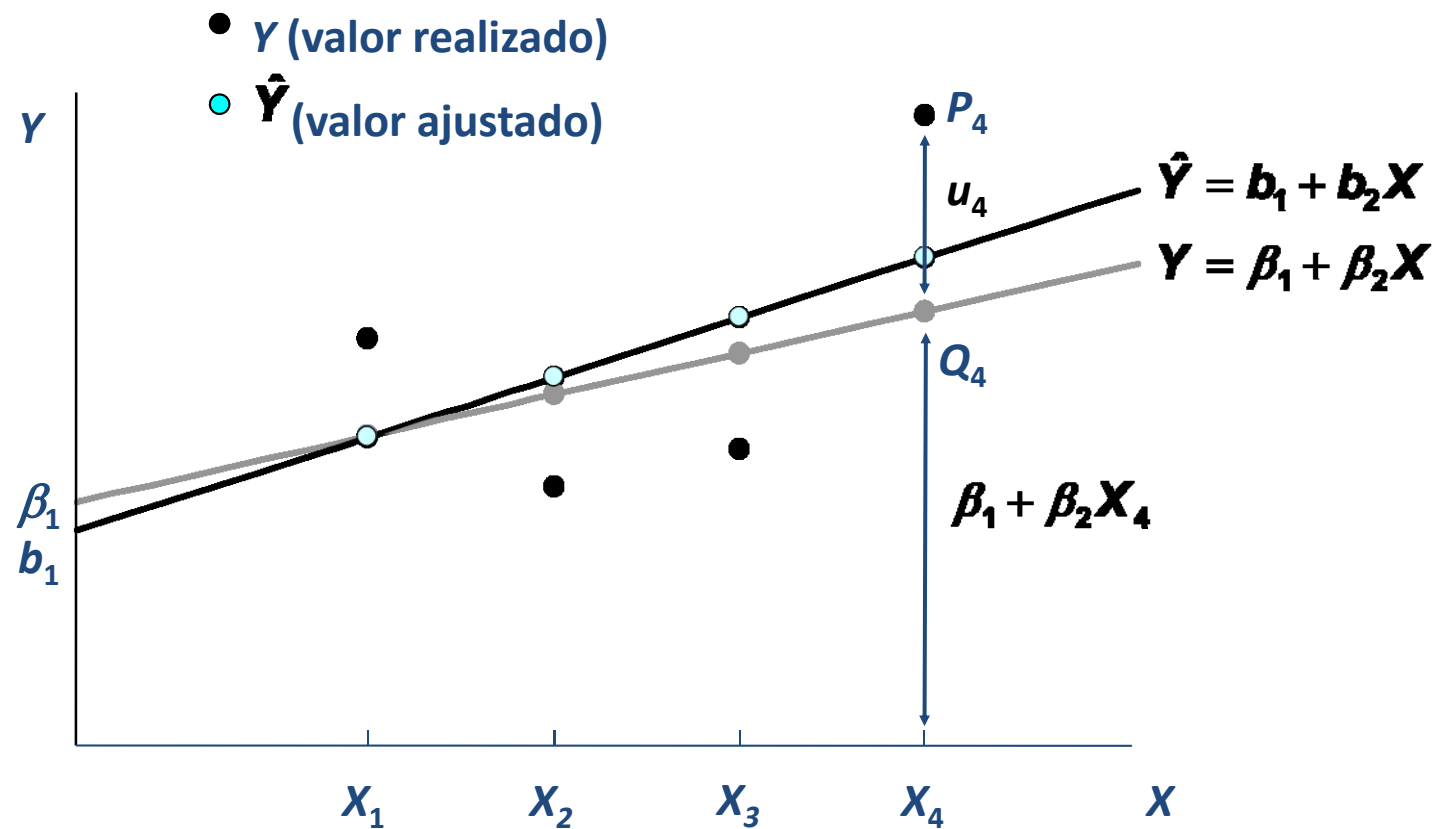
The residuals are the discrepancies between the actual and the fitted values.

Modelo de Regresión Simple



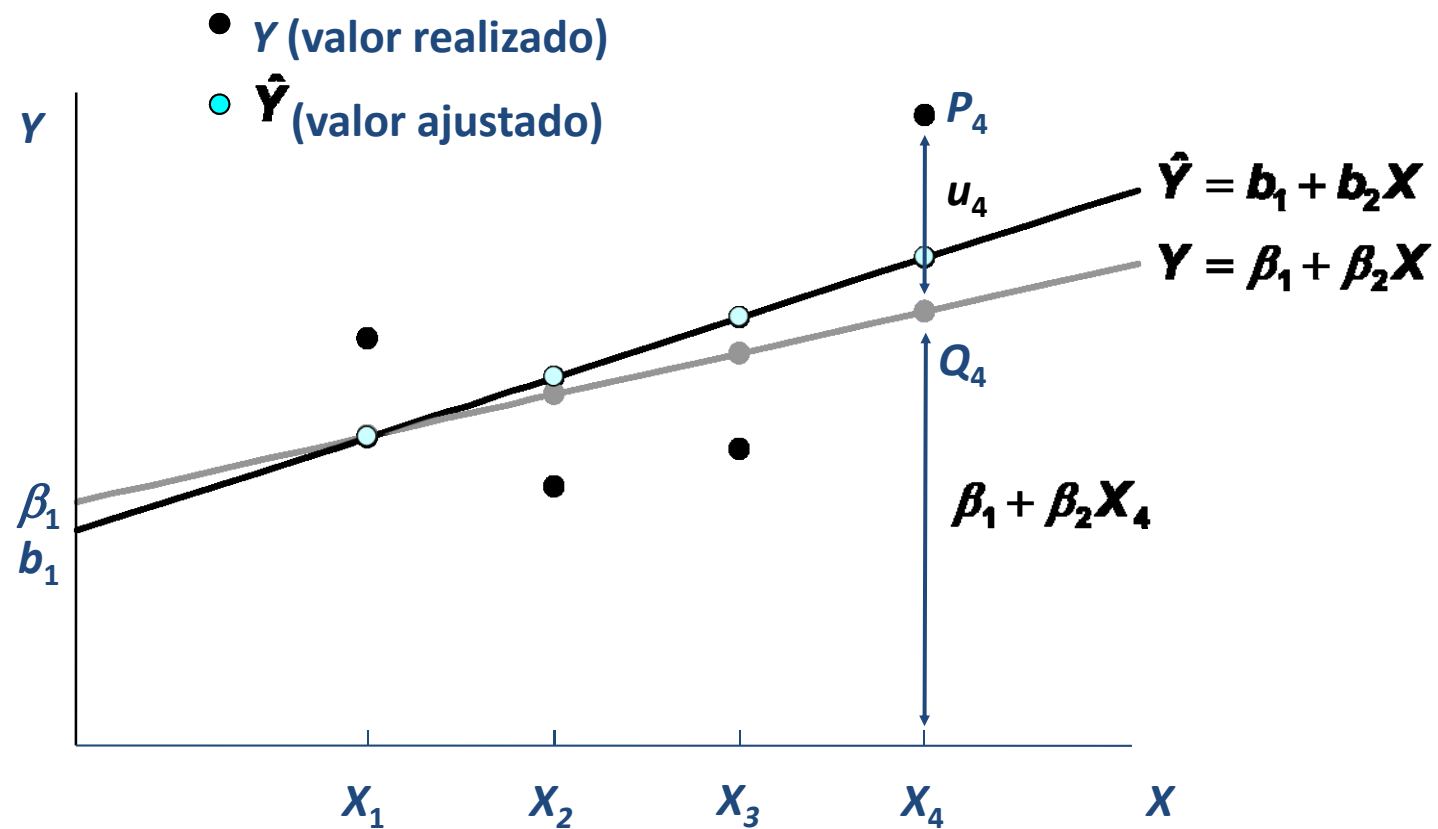
If the fit is a good one, the residuals and the values of the disturbance term will be similar, but they must be kept apart conceptually.

Modelo de Regresión Simple



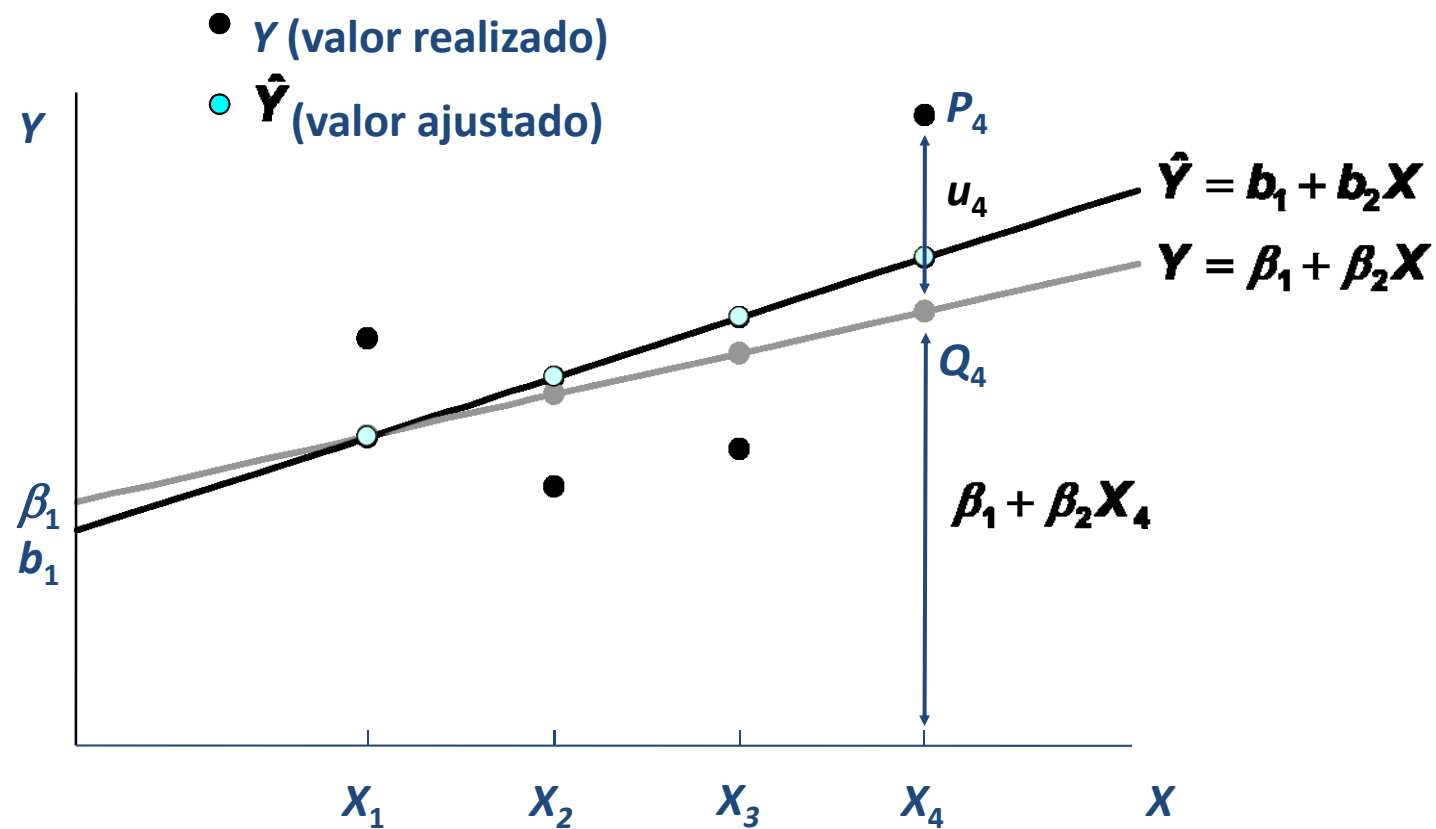
Both of these lines will be used in our analysis. Each permits a decomposition of the value of Y . The decompositions will be illustrated with the fourth observation.

Modelo de Regresión Simple



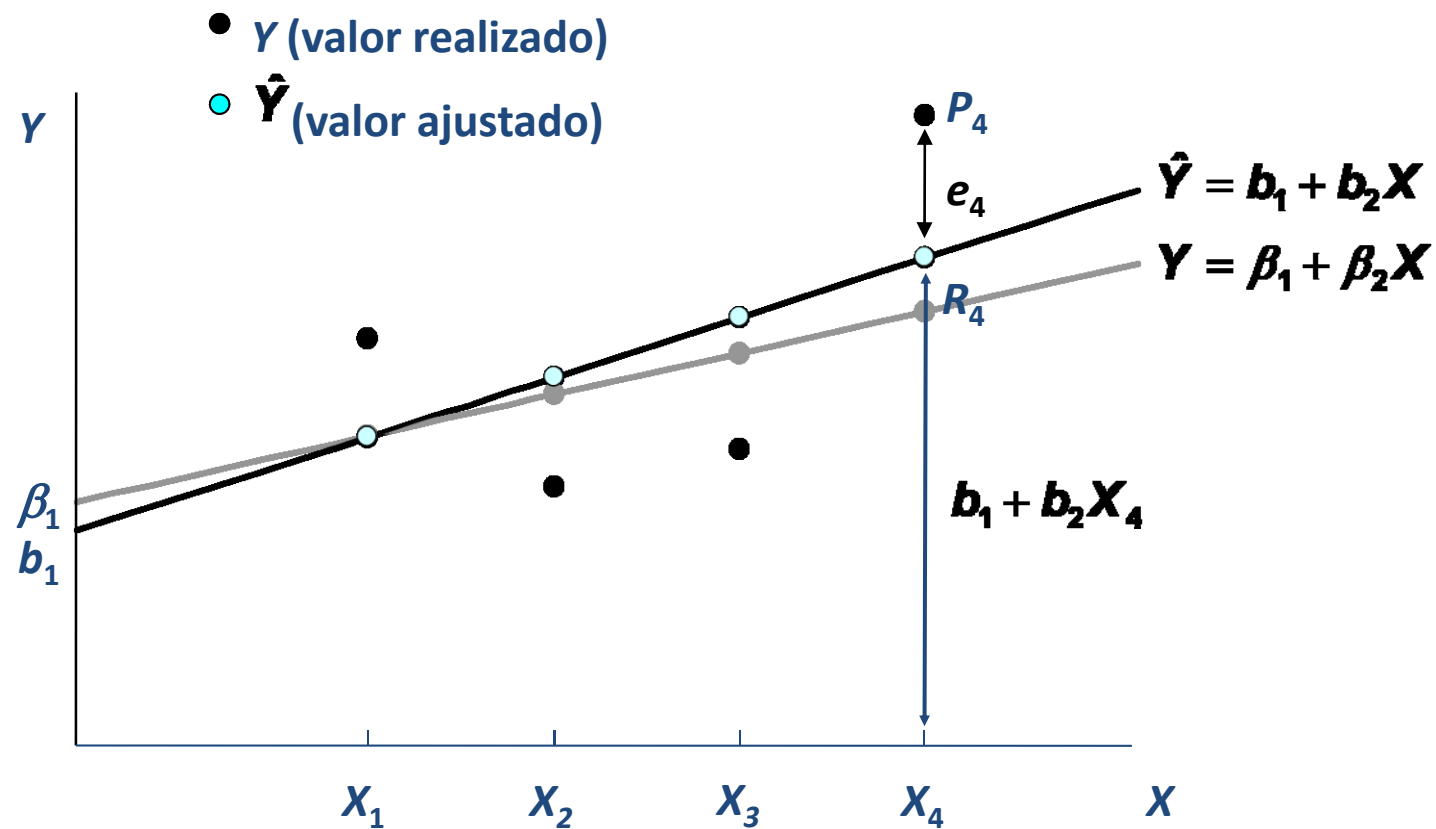
Using the theoretical relationship, Y can be decomposed into its nonstochastic component $\beta_1 + \beta_2 X$ and its random component u .

Modelo de Regresión Simple



This is a theoretical decomposition because we do not know the values of β_1 or β_2 , or the values of the disturbance term. We shall use it in our analysis of the properties of the regression coefficients

Modelo de Regresión Simple



The other decomposition is with reference to the fitted line. In each observation, the actual value of Y is equal to the fitted value plus the residual. This is an operational decomposition which we will use for practical purposes.

Criterio de Mínimos Cuadrados

Minimizar RSS (suma de errores al cuadrado)

$$RSS = \sum_{i=1}^n e_i^2 = e_1^2 + \dots + e_n^2$$

Criterio de Mínimos Cuadrados

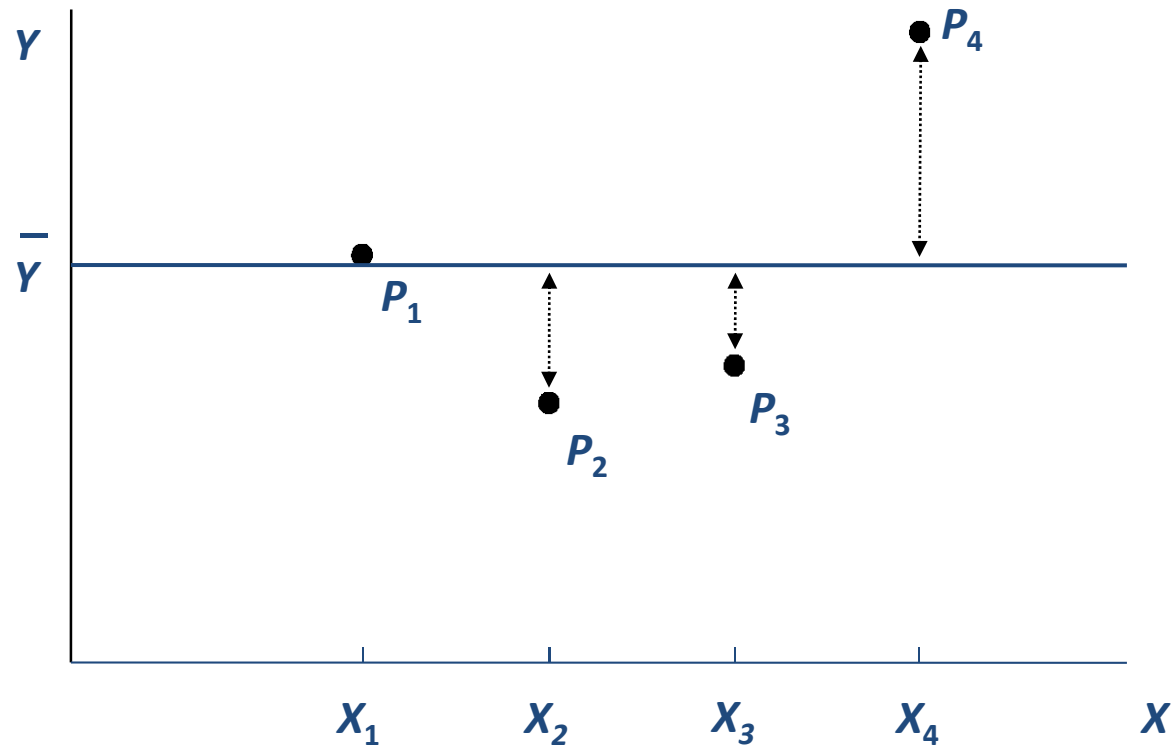
Minimizar RSS (suma de errores al cuadrado)

$$RSS = \sum_{i=1}^n e_i^2 = e_1^2 + \dots + e_n^2$$

Por qué no minimizar

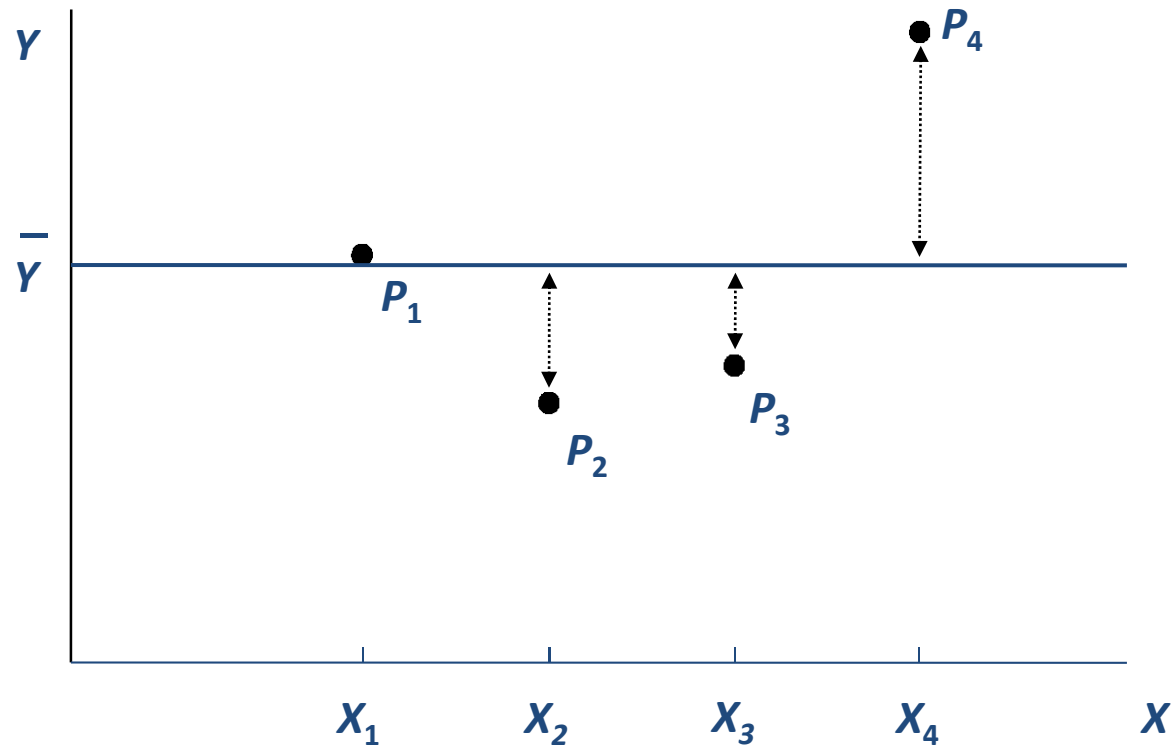
$$\sum_{i=1}^n e_i = e_1 + \dots + e_n$$

Modelo de Regresión Simple



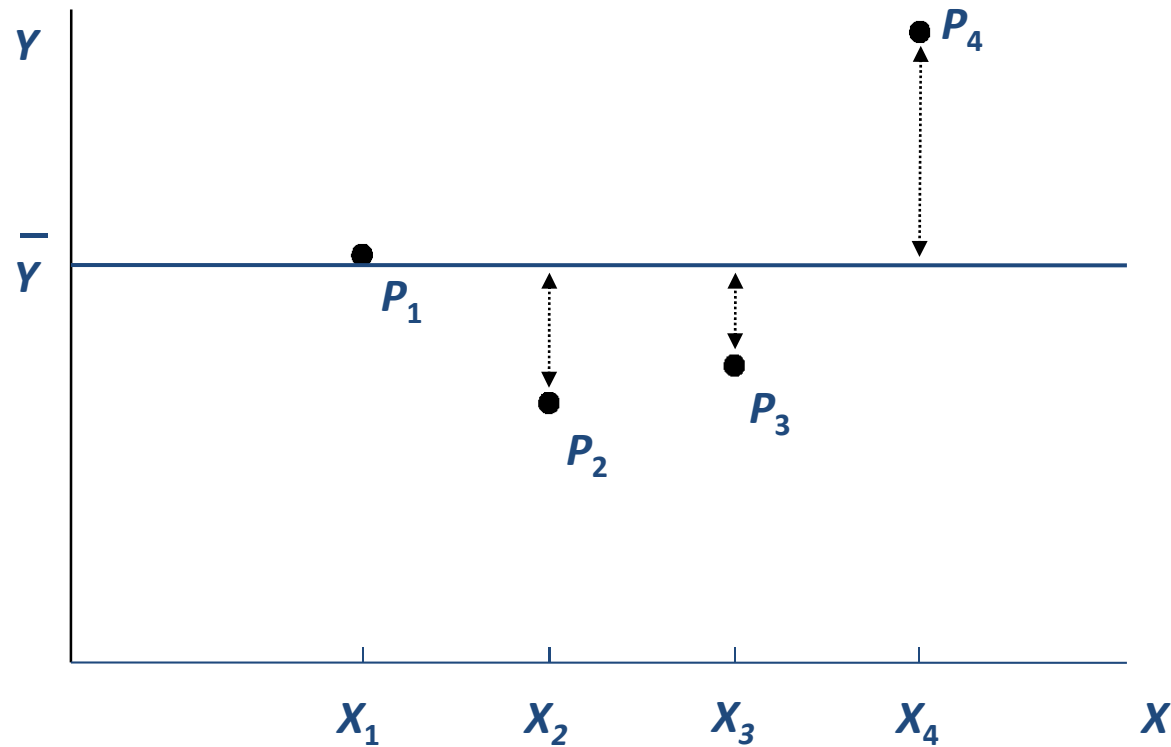
The answer is that you would get an apparently perfect fit by drawing a horizontal line through the mean value of Y . The sum of the residuals would be zero.

Modelo de Regresión Simple



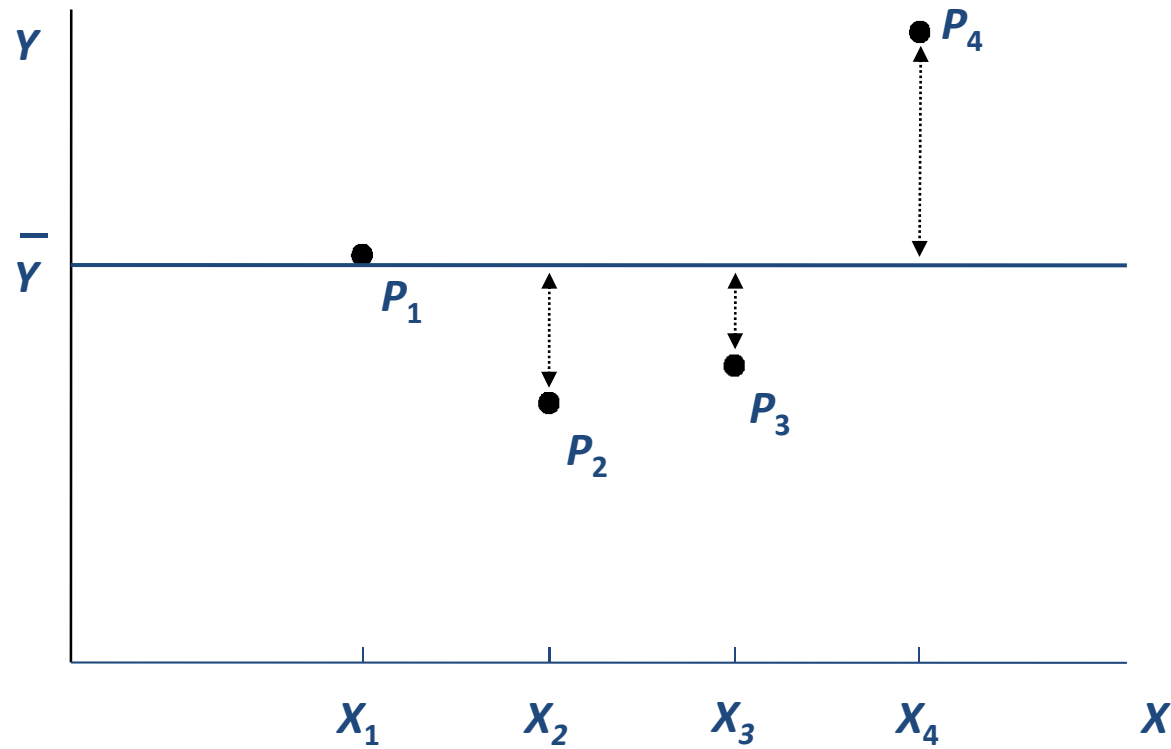
The other decomposition is with reference to the fitted line. In each observation, the actual value of Y is equal to the fitted value plus the residual. This is an operational decomposition which we will use for practical purposes.

Modelo de Regresión Simple



Of course there are other ways of dealing with the problem. The least squares criterion has the attraction that the estimators derived with it have desirable properties, provided that certain conditions are satisfied.

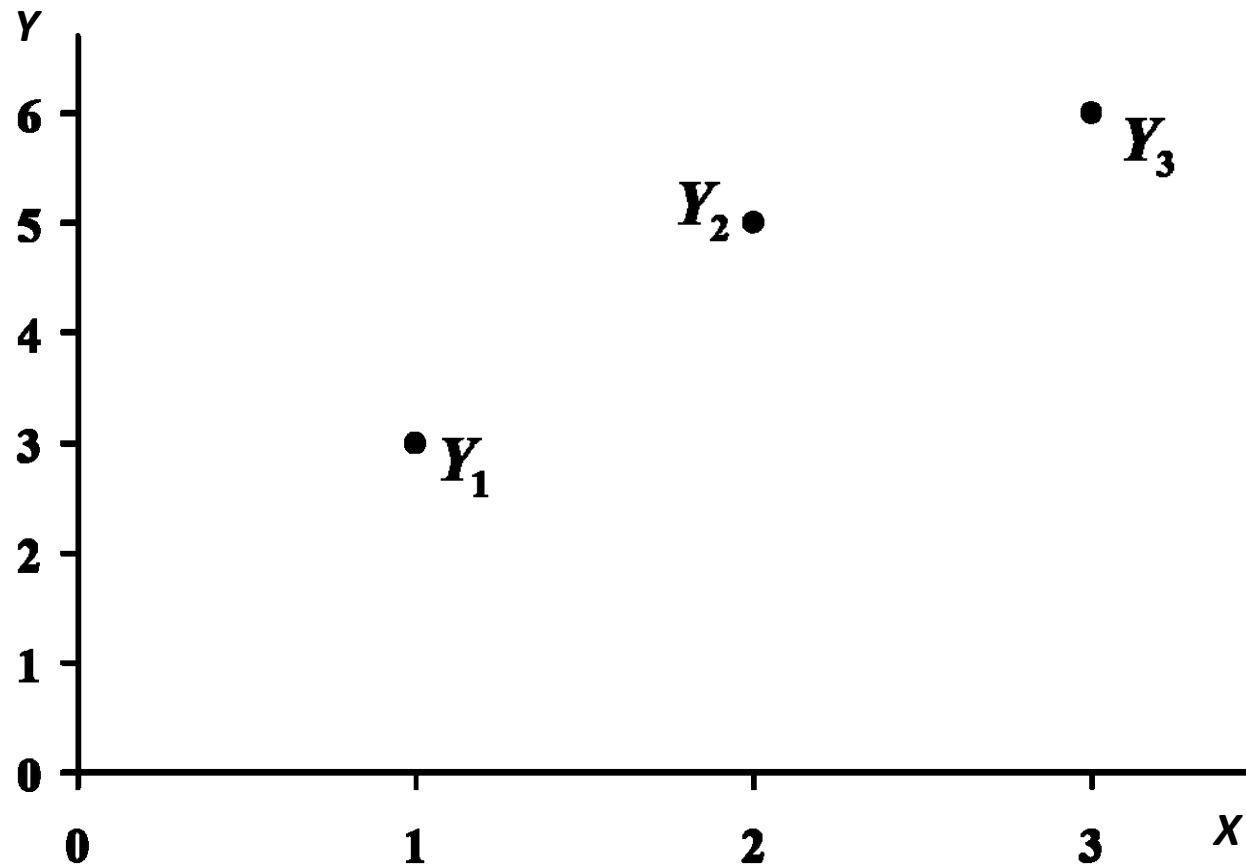
Modelo de Regresión Simple



The next sequence shows how the least squares criterion is used to calculate the coefficients of the fitted line.

DERIVING LINEAR REGRESSION COEFFICIENTS

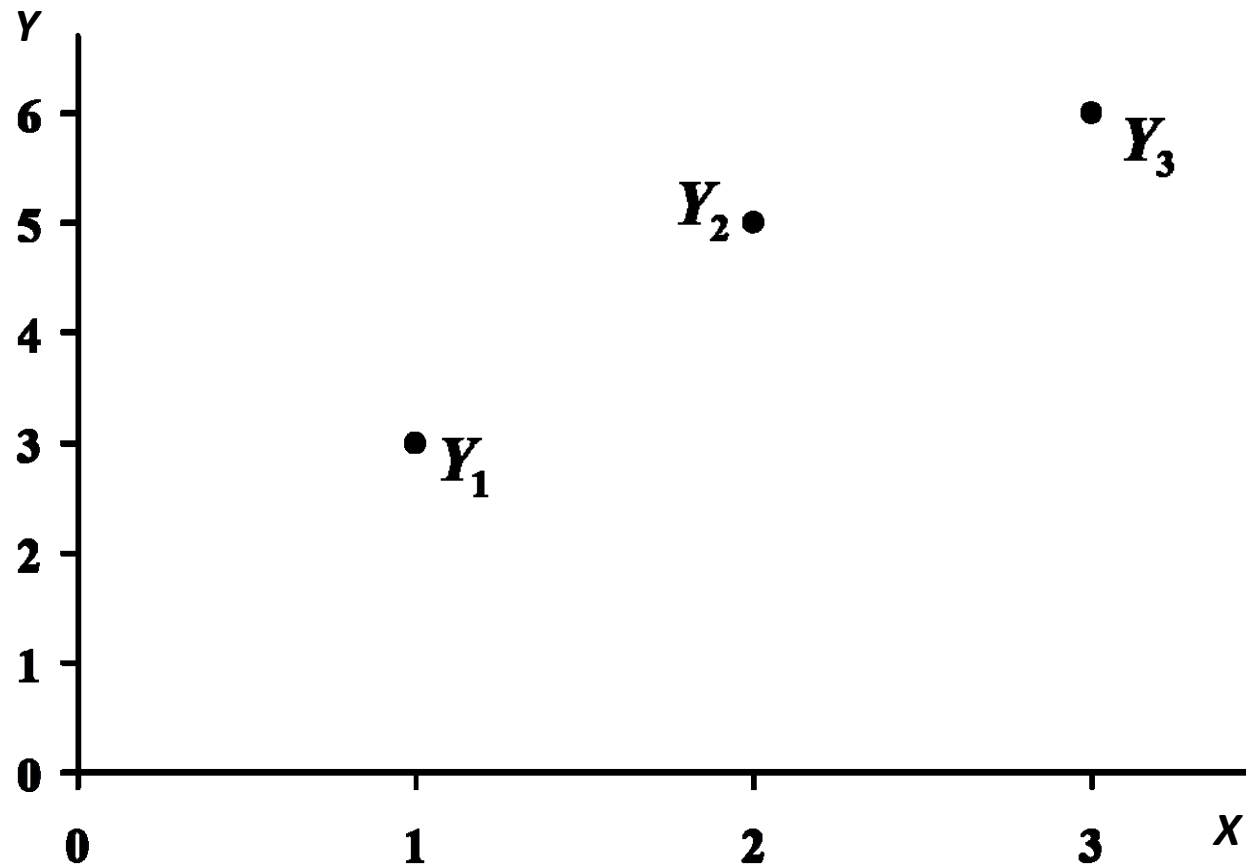
$$\text{True model: } Y = \beta_1 + \beta_2 X + u$$



This sequence shows how the regression coefficients for a simple regression model are derived, using the least squares criterion (OLS, for ordinary least squares)

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\text{True model: } Y = \beta_1 + \beta_2 X + u$$

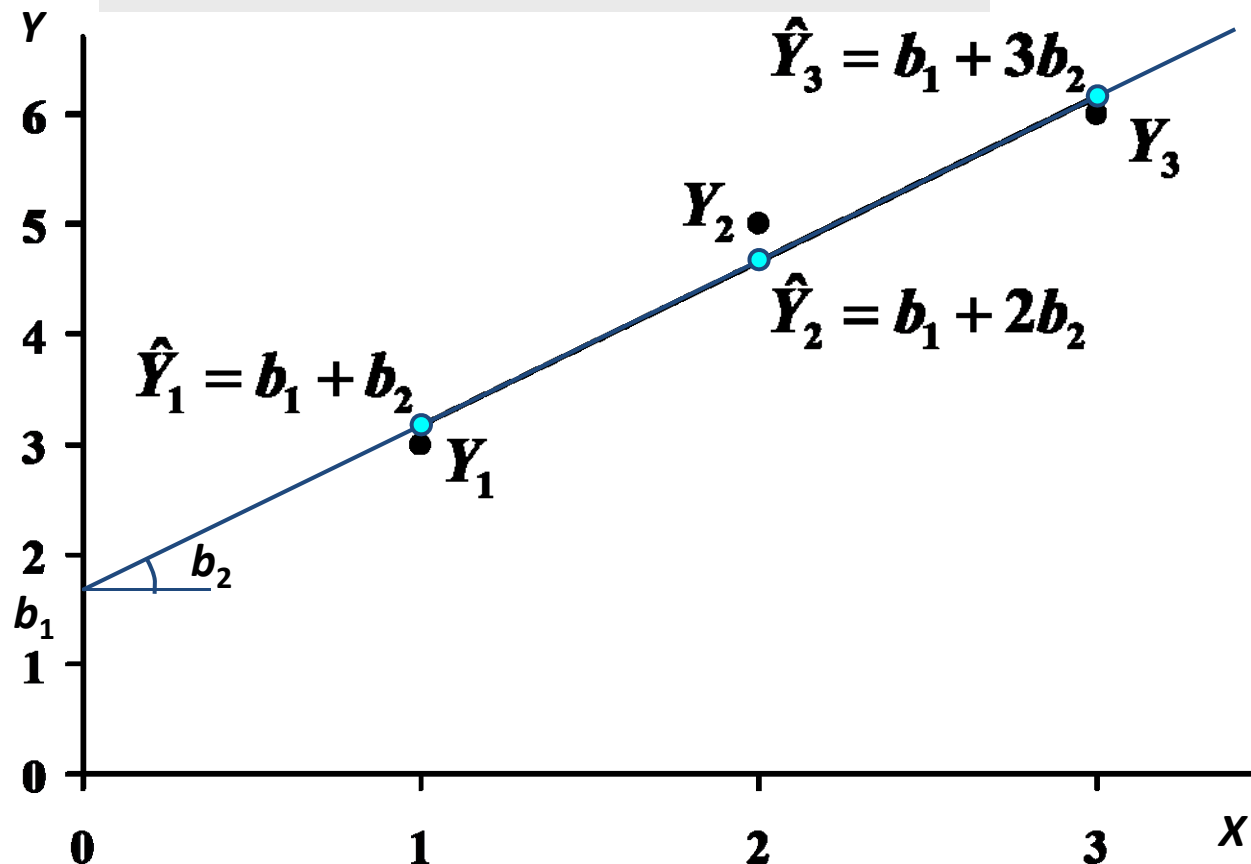


We will start with a numerical example with just three observations: (1,3), (2,5), and (3,6).

DERIVING LINEAR REGRESSION COEFFICIENTS

True model : $Y = \beta_1 + \beta_2 X + u$

Fitted line : $\hat{Y} = b_1 + b_2 X$

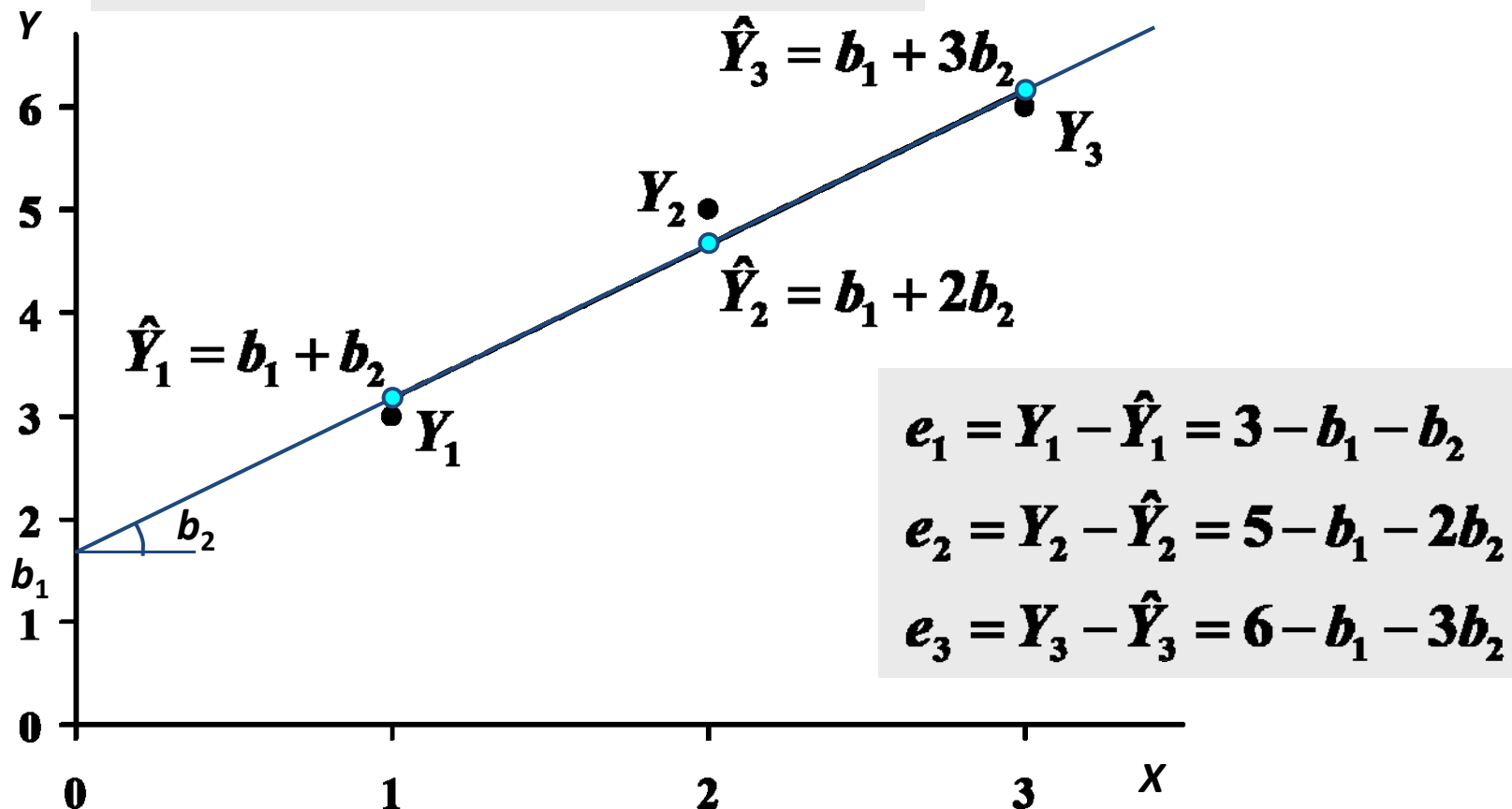


Writing the fitted regression as $\hat{Y} = b_1 + b_2 X$, we will determine the values of b_1 and b_2 that minimize RSS , the sum of the squares of the residuals.

DERIVING LINEAR REGRESSION COEFFICIENTS

True model : $Y = \beta_1 + \beta_2 X + u$

Fitted line : $\hat{Y} = b_1 + b_2 X$



Given our choice of b_1 and b_2 , the residuals are as shown.

SIMPLE REGRESSION ANALYSIS

$$RSS = e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2$$

$$\begin{aligned} e_1 &= Y_1 - \hat{Y}_1 = 3 - b_1 - b_2 \\ e_2 &= Y_2 - \hat{Y}_2 = 5 - b_1 - 2b_2 \\ e_3 &= Y_3 - \hat{Y}_3 = 6 - b_1 - 3b_2 \end{aligned}$$

The sum of the squares of the residuals is thus as shown above.

SIMPLE REGRESSION ANALYSIS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2\end{aligned}$$

The quadratics have been expanded.

SIMPLE REGRESSION ANALYSIS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

Like terms have been added together.

SIMPLE REGRESSION ANALYSIS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

For a minimum, the partial derivatives of RSS with respect to b_1 and b_2 should be zero. (We should also check a second-order condition.)

SIMPLE REGRESSION ANALYSIS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

The first-order conditions give us two equations in two unknowns.

SIMPLE REGRESSION ANALYSIS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

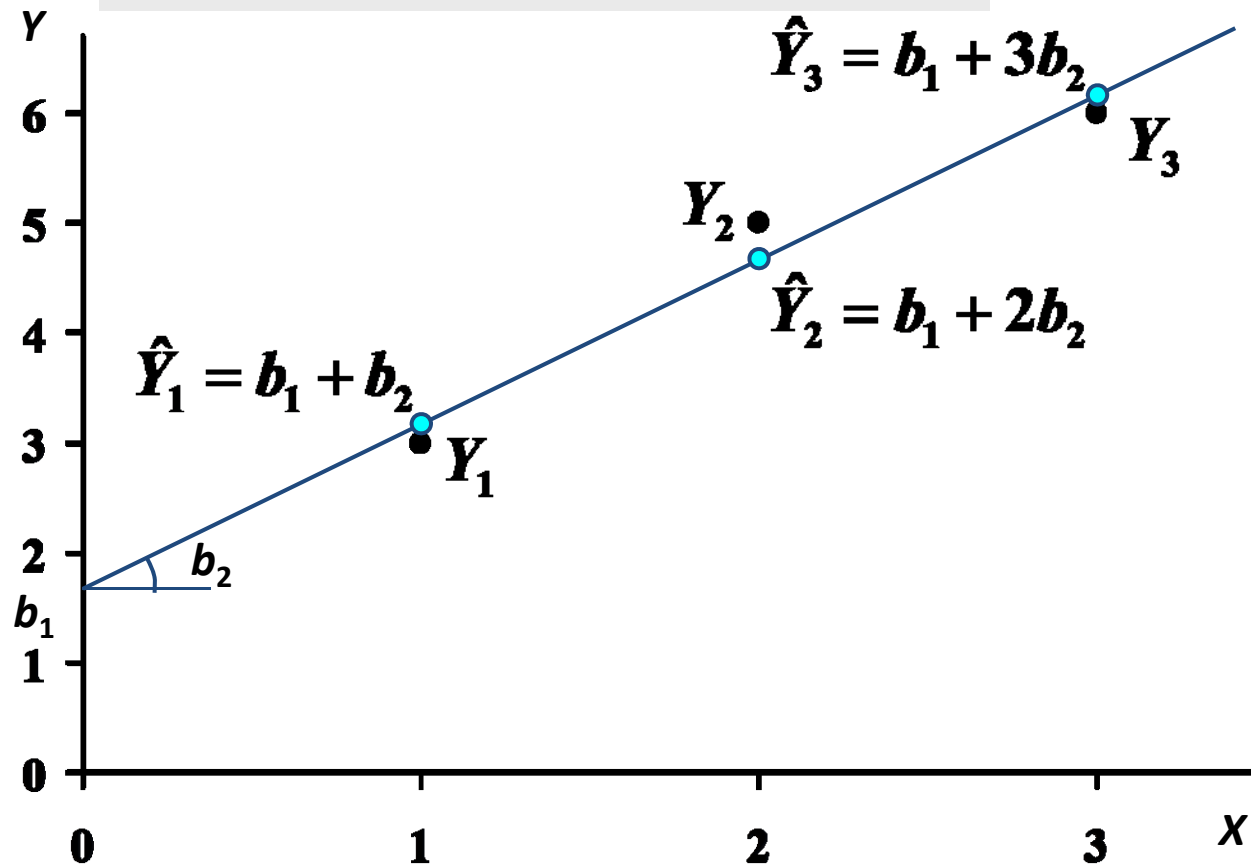
$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

Solving them, we find that RSS is minimized when b_1 and b_2 are equal to 1.67 and 1.50, respectively.

DERIVING LINEAR REGRESSION COEFFICIENTS

True model : $Y = \beta_1 + \beta_2 X + u$

Fitted line : $\hat{Y} = b_1 + b_2 X$

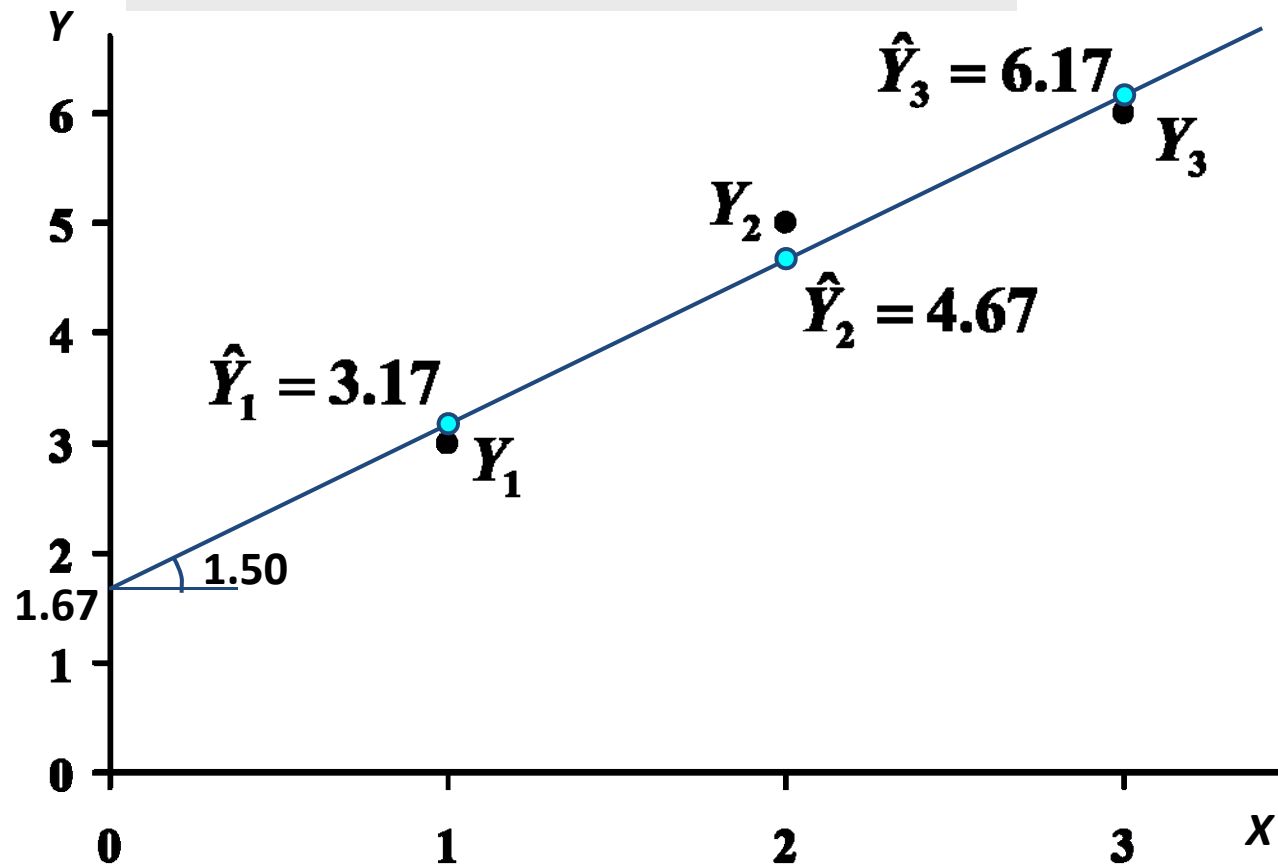


Here is the scatter diagram again.

DERIVING LINEAR REGRESSION COEFFICIENTS

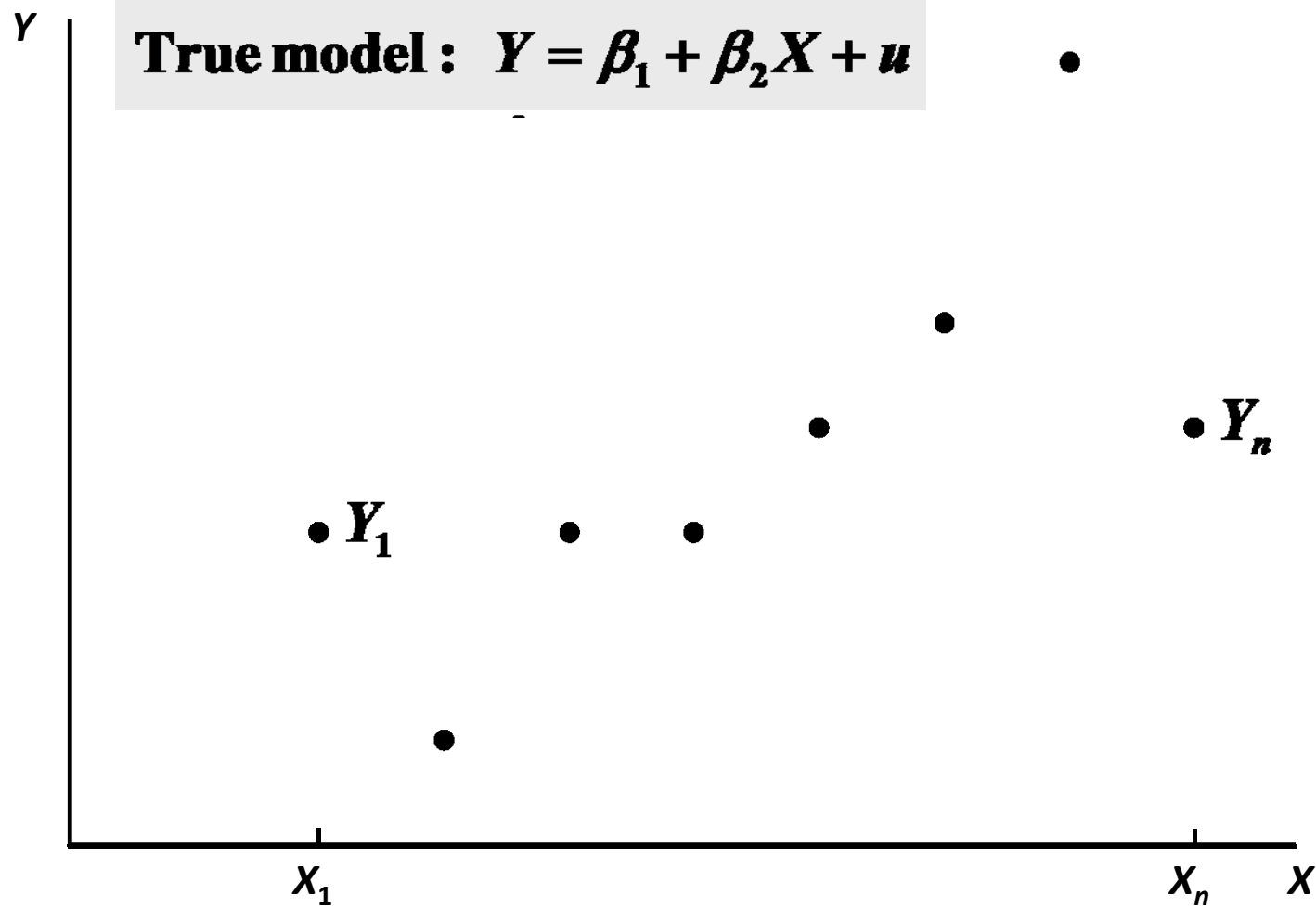
True model : $Y = \beta_1 + \beta_2 X + u$

Fitted line : $\hat{Y} = 1.67 + 1.50X$



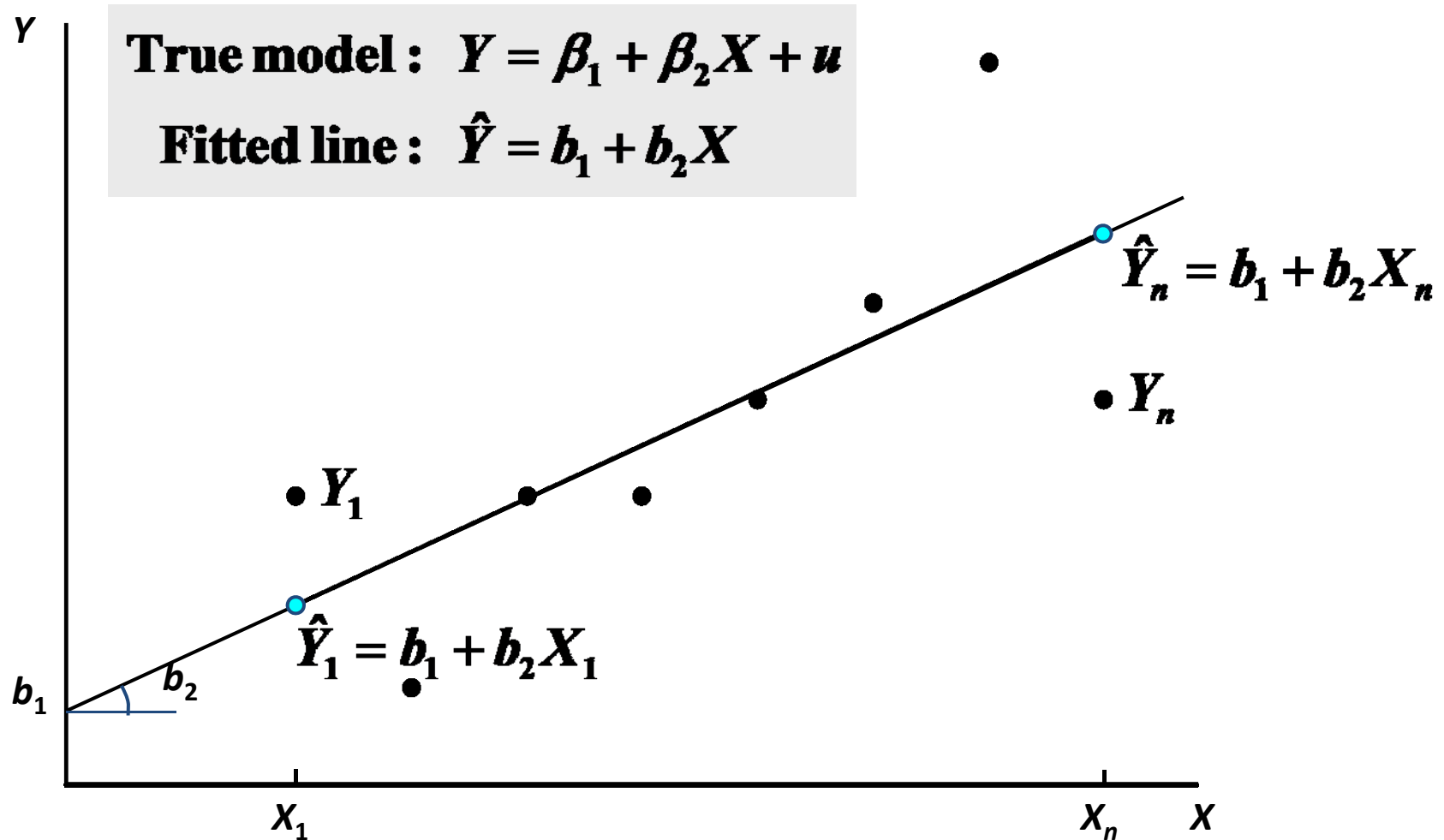
The fitted line and the fitted values of Y are as shown.

DERIVING LINEAR REGRESSION COEFFICIENTS



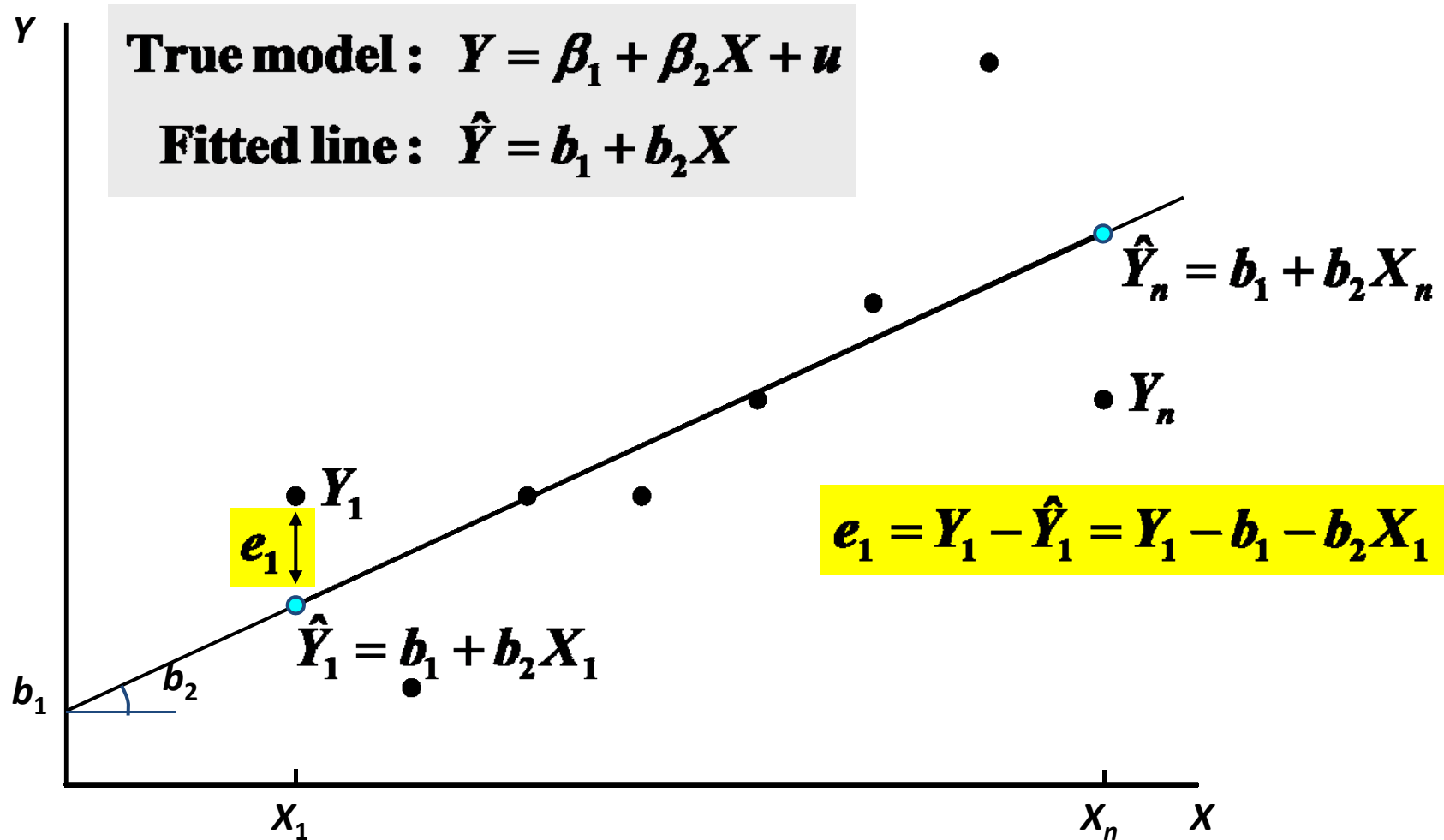
Now we will do the same thing for the general case with n observations.

DERIVING LINEAR REGRESSION COEFFICIENTS



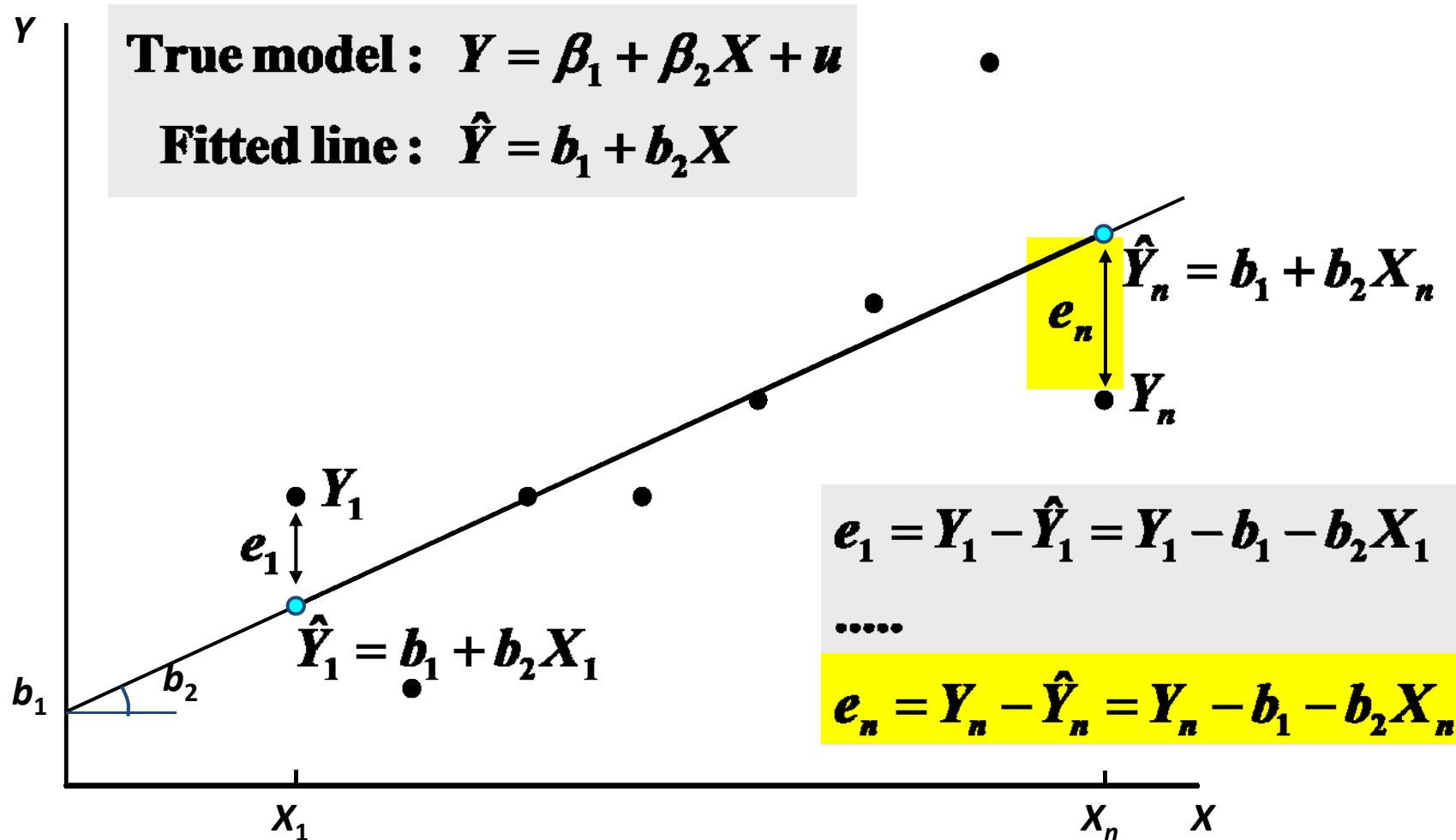
Given our choice of b_1 and b_2 , we will obtain a fitted line as shown.

DERIVING LINEAR REGRESSION COEFFICIENTS



The residual for the first observation is defined.

DERIVING LINEAR REGRESSION COEFFICIENTS



Similarly we define the residuals for the remaining observations. That for the last one is marked.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

$$RSS = e_1^2 + \dots + e_n^2 = (Y_1 - b_1 - b_2X_1)^2 + \dots + (Y_n - b_1 - b_2X_n)^2$$

RSS, the sum of the squares of the residuals, is defined for the general case. The data for the numerical example are shown for comparison..

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

$$\begin{aligned}RSS &= e_1^2 + \dots + e_n^2 = (Y_1 - b_1 - b_2X_1)^2 + \dots + (Y_n - b_1 - b_2X_n)^2 \\&= Y_1^2 + b_1^2 + b_2^2X_1^2 - 2b_1Y_1 - 2b_2X_1Y_1 + 2b_1b_2X_1 \\&\quad + \dots \\&\quad + Y_n^2 + b_1^2 + b_2^2X_n^2 - 2b_1Y_n - 2b_2X_nY_n + 2b_1b_2X_n\end{aligned}$$

The quadratics are expanded.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\begin{aligned}RSS &= e_1^2 + e_2^2 + e_3^2 = (3 - b_1 - b_2)^2 + (5 - b_1 - 2b_2)^2 + (6 - b_1 - 3b_2)^2 \\&= 9 + b_1^2 + b_2^2 - 6b_1 - 6b_2 + 2b_1b_2 \\&\quad + 25 + b_1^2 + 4b_2^2 - 10b_1 - 20b_2 + 4b_1b_2 \\&\quad + 36 + b_1^2 + 9b_2^2 - 12b_1 - 36b_2 + 6b_1b_2 \\&= 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2\end{aligned}$$

$$\begin{aligned}RSS &= e_1^2 + \dots + e_n^2 = (Y_1 - b_1 - b_2X_1)^2 + \dots + (Y_n - b_1 - b_2X_n)^2 \\&= Y_1^2 + b_1^2 + b_2^2X_1^2 - 2b_1Y_1 - 2b_2X_1Y_1 + 2b_1b_2X_1 \\&\quad + \dots \\&\quad + Y_n^2 + b_1^2 + b_2^2X_n^2 - 2b_1Y_n - 2b_2X_nY_n + 2b_1b_2X_n \\&= \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_iY_i + 2b_1b_2 \sum X_i\end{aligned}$$

Like terms are added together.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$RSS = 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

Note that in this equation the observations on X and Y are just data that determine the coefficients in the expression for RSS .

DERIVING LINEAR REGRESSION COEFFICIENTS

$$RSS = 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

The choice variables in the expression are b_1 and b_2 . This may seem a bit strange because in elementary calculus courses b_1 and b_2 are usually constants and X and Y are variables.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$RSS = 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

However, if you have any doubts, compare what we are doing in the general case with what we did in the numerical example.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$RSS = 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 2nb_1 - 2\sum Y_i + 2b_2 \sum X_i = 0$$

The first derivative with respect to b_1 .

DERIVING LINEAR REGRESSION COEFFICIENTS

$$RSS = 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 2nb_1 - 2 \sum Y_i + 2b_2 \sum X_i = 0$$

$$nb_1 = \sum Y_i - b_2 \sum X_i \quad b_1 = \bar{Y} - b_2 \bar{X}$$

With some simple manipulation we obtain a tidy expression for b_1 .

DERIVING LINEAR REGRESSION COEFFICIENTS

$$RSS = 70 + 3b_1^2 + 14b_2^2 - 28b_1 - 62b_2 + 12b_1b_2$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 6b_1 + 12b_2 - 28 = 0$$

$$\therefore b_1 = 1.67, \quad b_2 = 1.50$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 12b_1 + 28b_2 - 62 = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 2nb_1 - 2 \sum Y_i + 2b_2 \sum X_i = 0$$

$$nb_1 = \sum Y_i - b_2 \sum X_i \quad b_1 = \bar{Y} - b_2 \bar{X}$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

The first derivative with respect to b_2 .

SIMPLE REGRESSION ANALYSIS

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + b_1 \sum X_i = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 2nb_1 - 2 \sum Y_i + 2b_2 \sum X_i = 0$$

$$nb_1 = \sum Y_i - b_2 \sum X_i \quad b_1 = \bar{Y} - b_2 \bar{X}$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

Divide through by 2.

SIMPLE REGRESSION ANALYSIS

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) \sum X_i = 0$$

$$RSS = \sum Y_i^2 + nb_1^2 + b_2^2 \sum X_i^2 - 2b_1 \sum Y_i - 2b_2 \sum X_i Y_i + 2b_1 b_2 \sum X_i$$

$$\frac{\partial RSS}{\partial b_1} = 0 \Rightarrow 2nb_1 - 2 \sum Y_i + 2b_2 \sum X_i = 0$$

$$nb_1 = \sum Y_i - b_2 \sum X_i \quad b_1 = \bar{Y} - b_2 \bar{X}$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

We now substitute for b_1 using the expression obtained for it and we thus obtain an equation that contains b_2 only.

SIMPLE REGRESSION ANALYSIS

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) n \bar{X} = 0$$

$$\bar{X} = \frac{\sum X_i}{n}$$

$$\sum X_i = n \bar{X}$$

The definition of the sample mean has been used.

SIMPLE REGRESSION ANALYSIS

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) n \bar{X} = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + n \bar{X} \bar{Y} - n b_2 \bar{X}^2 = 0$$

The last two terms have been disentangled.

SIMPLE REGRESSION ANALYSIS

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + b_1 \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) \sum X_i = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + (\bar{Y} - b_2 \bar{X}) n \bar{X} = 0$$

$$b_2 \sum X_i^2 - \sum X_i Y_i + n \bar{X} \bar{Y} - n b_2 \bar{X}^2 = 0$$

$$b_2 (\sum X_i^2 - n \bar{X}^2) = \sum X_i Y_i - n \bar{X} \bar{Y}$$

Terms not involving b_2 have been transferred to the right side.

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

To create space, the equation is shifted to the top of the slide.

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

Hence we obtain an expression for b_2 .

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

$$b_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

In practice, we shall use an alternative expression. We will demonstrate that it is equivalent.

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

$$b_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = \sum X_i Y_i - \sum X_i \bar{Y} - \sum \bar{X} Y_i + \sum \bar{X} \bar{Y}$$

Expanding the numerator, we obtain the terms shown.

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

$$b_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\begin{aligned}\sum (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum X_i Y_i - \sum X_i \bar{Y} - \sum \bar{X} Y_i + \sum \bar{X} \bar{Y} \\ &= \sum X_i Y_i - \bar{Y} \sum X_i - \bar{X} \sum Y_i + n\bar{X}\bar{Y}\end{aligned}$$

In the second term the mean value of Y is a common factor. In the third, the mean value of X is a common factor. The last term is the same for all i .

SIMPLE REGRESSION ANALYSIS

$$\bar{X} = \frac{\sum X_i}{n}$$

$$\sum X_i = n\bar{X}$$

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

$$b_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\begin{aligned}\sum (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum X_i Y_i - \sum X_i \bar{Y} - \sum \bar{X} Y_i + \sum \bar{X} \bar{Y} \\ &= \sum X_i Y_i - \bar{Y} \sum X_i - \bar{X} \sum Y_i + n\bar{X}\bar{Y} \\ &= \sum X_i Y_i - \bar{Y}(n\bar{X}) - \bar{X}(n\bar{Y}) + n\bar{X}\bar{Y}\end{aligned}$$

We use the definitions of the sample means to simplify the expression.

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

$$b_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\begin{aligned}\sum (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum X_i Y_i - \sum X_i \bar{Y} - \sum \bar{X} Y_i + \sum \bar{X} \bar{Y} \\ &= \sum X_i Y_i - \bar{Y} \sum X_i - \bar{X} \sum Y_i + n\bar{X}\bar{Y} \\ &= \sum X_i Y_i - \bar{Y}(n\bar{X}) - \bar{X}(n\bar{Y}) + n\bar{X}\bar{Y} \\ &= \sum X_i Y_i - n\bar{X}\bar{Y}\end{aligned}$$

Hence we have shown that the numerators of the two expressions are the same.

SIMPLE REGRESSION ANALYSIS

$$b_2(\sum X_i^2 - n\bar{X}^2) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$b_2 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

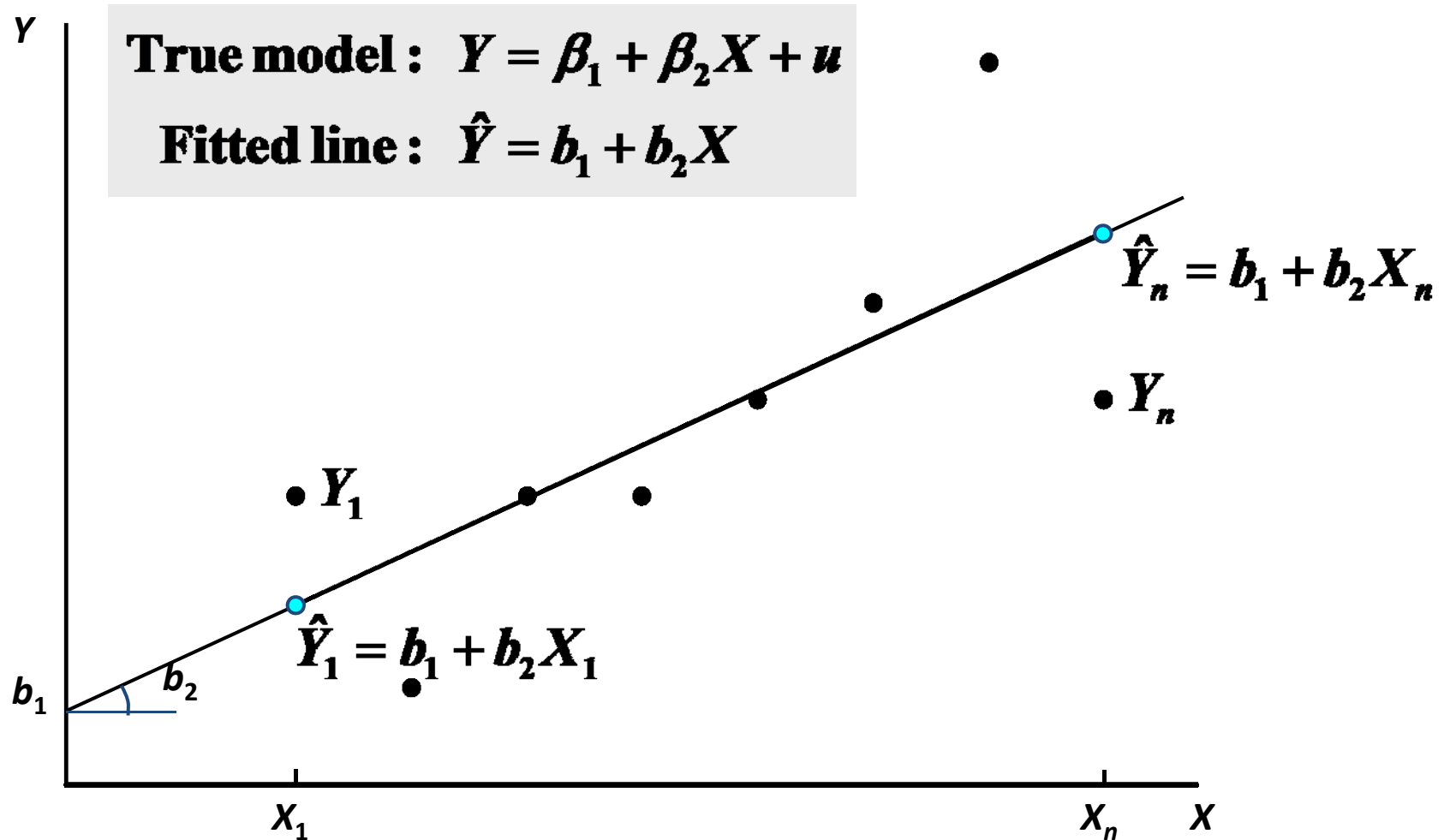
$$b_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = \sum X_i Y_i - n\bar{X}\bar{Y}$$

$$\sum (X_i - \bar{X})^2 = \sum X_i^2 - n\bar{X}^2$$

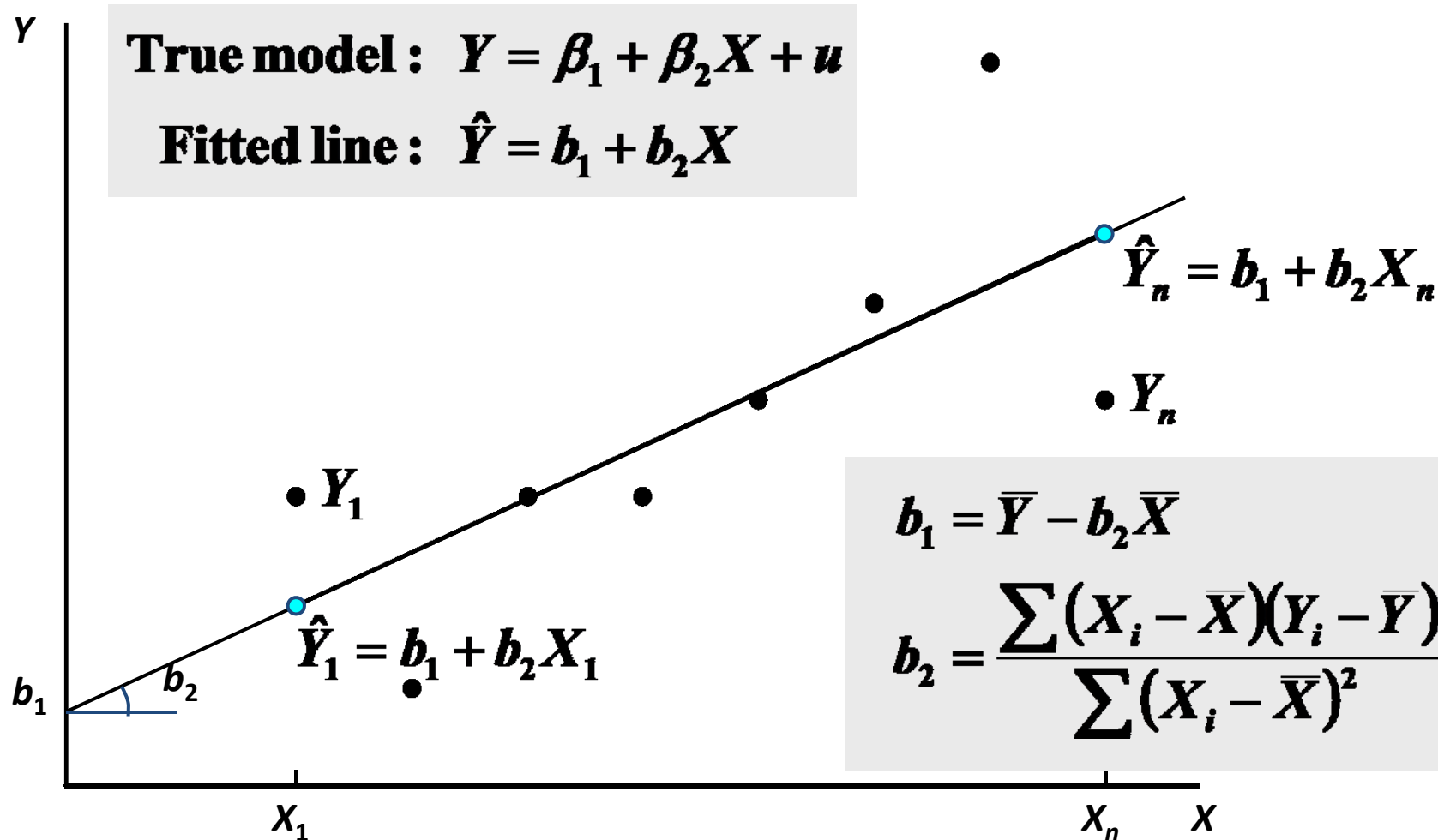
The denominator is mathematically a special case of the numerator, replacing Y by X . Hence the expressions are equivalent.

DERIVING LINEAR REGRESSION COEFFICIENTS



The scatter diagram is shown again. We will summarize what we have done. We hypothesized that the true model is as shown, we obtained some data, and we fitted a line.

DERIVING LINEAR REGRESSION COEFFICIENTS



We chose the parameters of the fitted line so as to minimize the sum of the squares of the residuals. As a result, we derived the expressions for b_1 and b_2 .

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\text{True model : } Y = \beta_2 X + u$$

$$\text{Fitted line : } \hat{Y} = b_2 X$$

Typically, an intercept should be included in the regression specification. Occasionally, however, one may have reason to fit the regression without an intercept. In the case of a simple regression model, the true and fitted models become as shown.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\text{True model : } Y = \beta_2 X + u$$

$$\text{Fitted line : } \hat{Y} = b_2 X$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_2 X_i$$

We will derive the expression for b_2 from first principles using the least squares criterion.
The residual in observation i is $e_i = Y_i - b_2 X_i$.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\text{True model : } Y = \beta_2 X + u$$

$$\text{Fitted line : } \hat{Y} = b_2 X$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_2 X_i$$

$$RSS = \sum_{i=1}^n (Y_i - b_2 X_i)^2 = \sum_{i=1}^n Y_i^2 - 2b_2 \sum_{i=1}^n X_i Y_i + b_2^2 \sum_{i=1}^n X_i^2$$

With this, we obtain the expression for the sum of the squares of the residuals.

DERIVING LINEAR REGRESSION COEFFICIENTS

True model : $Y = \beta_2 X + u$

Fitted line : $\hat{Y} = b_2 X$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_2 X_i$$

$$RSS = \sum_{i=1}^n (Y_i - b_2 X_i)^2 = \sum_{i=1}^n Y_i^2 - 2b_2 \sum_{i=1}^n X_i Y_i + b_2^2 \sum_{i=1}^n X_i^2$$

$$\frac{dRSS}{db_2} = 2b_2 \sum_{i=1}^n X_i^2 - 2 \sum_{i=1}^n X_i Y_i = 0$$

Differentiating with respect to b_2 , we obtain the first-order condition for a minimum.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\text{True model : } Y = \beta_2 X + u$$

$$\text{Fitted line : } \hat{Y} = b_2 X$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_2 X_i$$

$$RSS = \sum_{i=1}^n (Y_i - b_2 X_i)^2 = \sum_{i=1}^n Y_i^2 - 2b_2 \sum_{i=1}^n X_i Y_i + b_2^2 \sum_{i=1}^n X_i^2$$

$$\frac{dRSS}{db_2} = 2b_2 \sum_{i=1}^n X_i^2 - 2 \sum_{i=1}^n X_i Y_i = 0$$

$$b_2 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}$$

Hence, we obtain the OLS estimator of b_2 for this model.

DERIVING LINEAR REGRESSION COEFFICIENTS

$$\text{True model : } Y = \beta_2 X + u$$

$$\text{Fitted line : } \hat{Y} = b_2 X$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_2 X_i$$

$$RSS = \sum_{i=1}^n (Y_i - b_2 X_i)^2 = \sum_{i=1}^n Y_i^2 - 2b_2 \sum_{i=1}^n X_i Y_i + b_2^2 \sum_{i=1}^n X_i^2$$

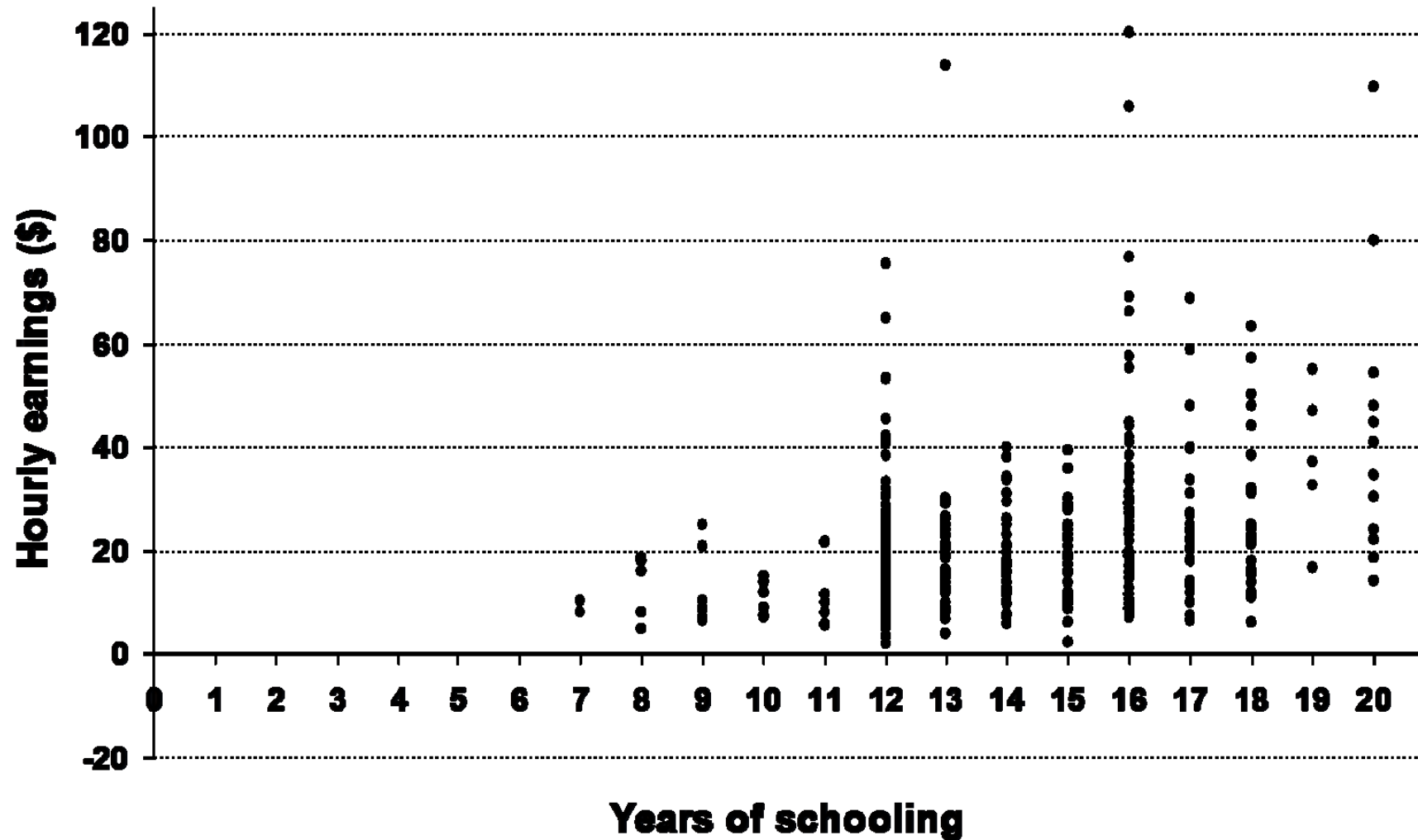
$$\frac{dRSS}{db_2} = 2b_2 \sum_{i=1}^n X_i^2 - 2 \sum_{i=1}^n X_i Y_i = 0$$

$$b_2 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}$$

$$\frac{d^2 RSS}{db_2^2} = 2 \sum_{i=1}^n X_i^2 > 0$$

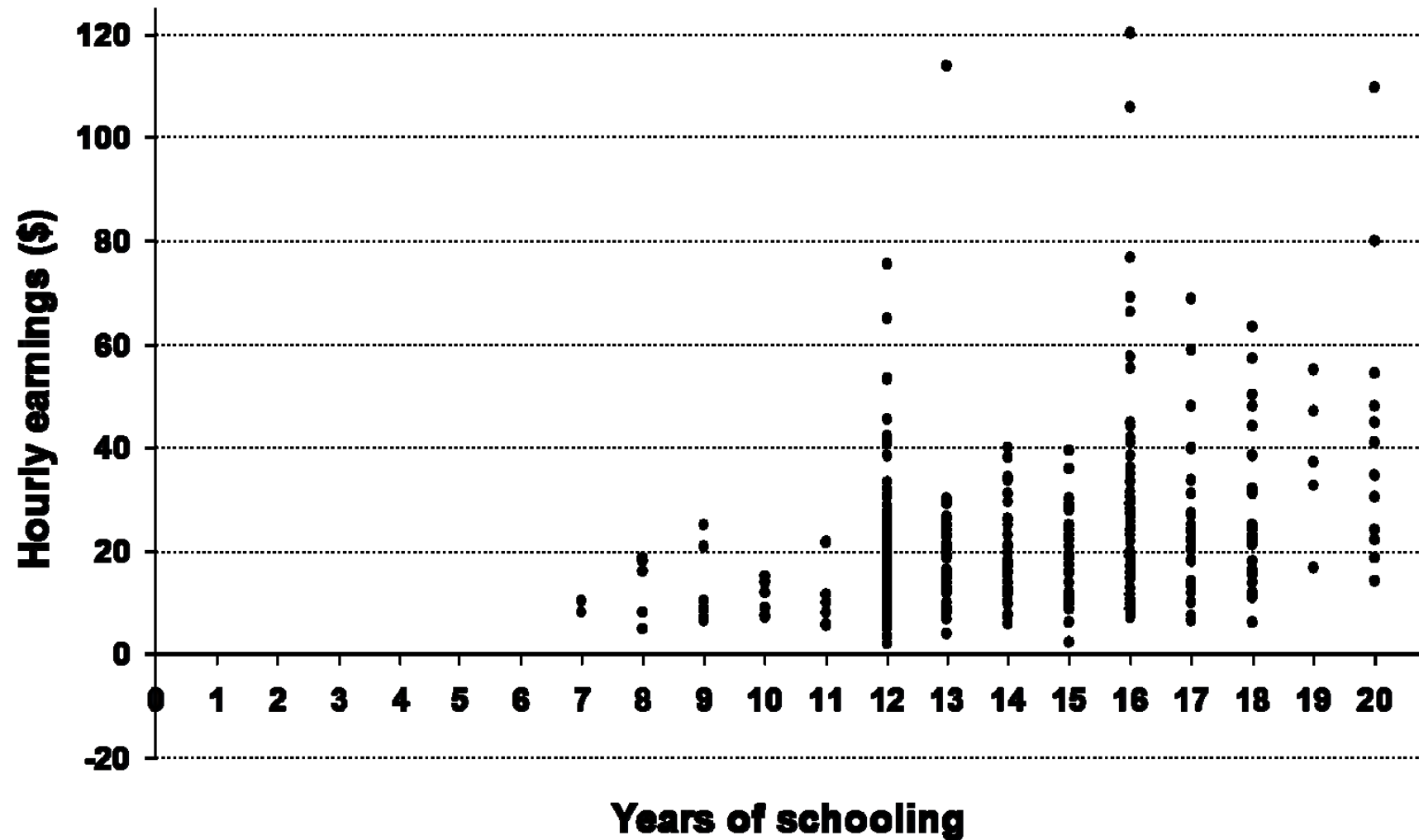
The second derivative is positive, confirming that we have found a minimum.

INTERPRETATION OF A REGRESSION EQUATION



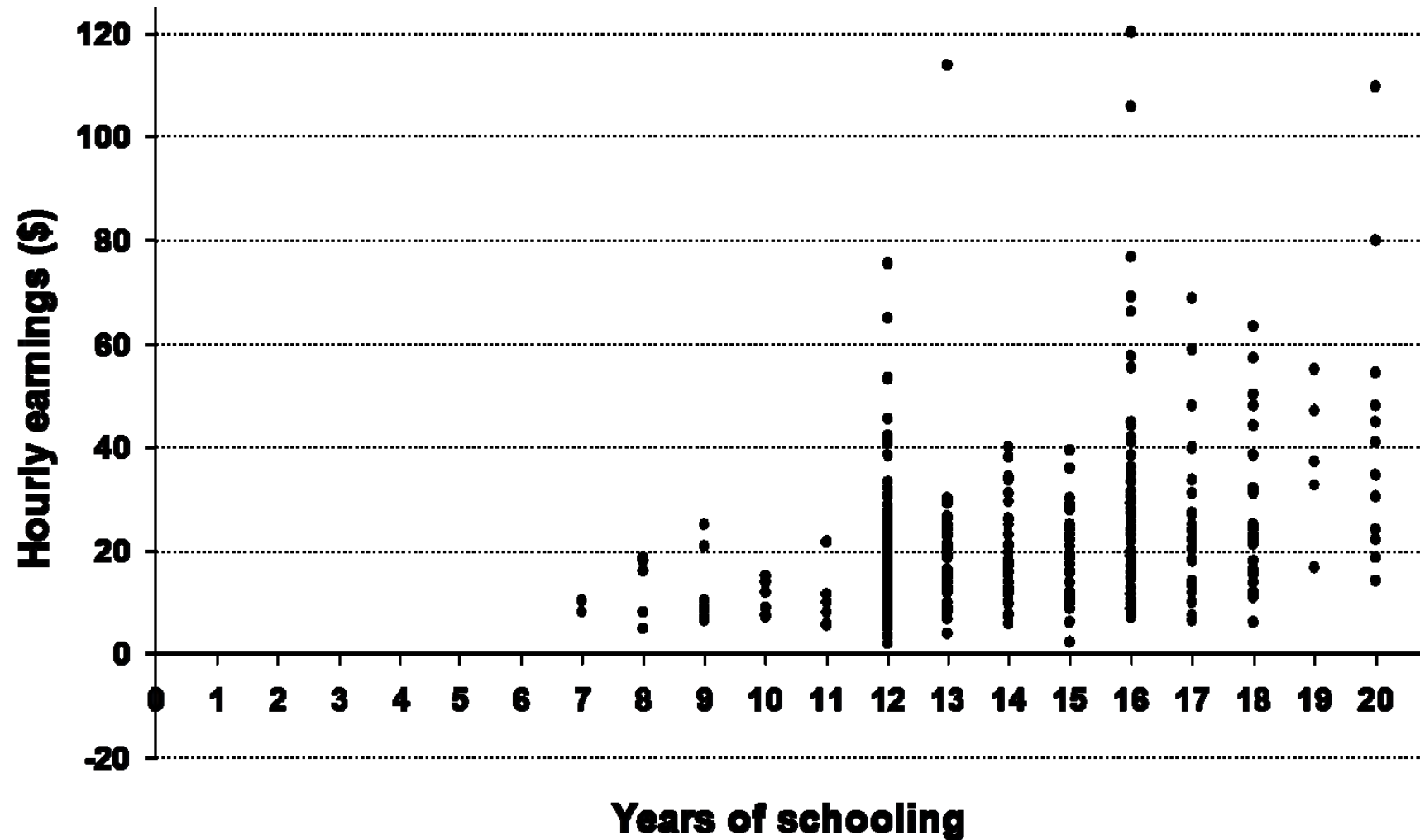
The scatter diagram shows hourly earnings in 2002 plotted against years of schooling, defined as highest grade completed, for a sample of 540 respondents from the National Longitudinal Survey of Youth.

INTERPRETATION OF A REGRESSION EQUATION



Highest grade completed means just that for elementary and high school. Grades 13, 14, and 15 mean completion of one, two and three years of college.

INTERPRETATION OF A REGRESSION EQUATION



Grade 16 means completion of four-year college. Higher grades indicate years of postgraduate education.

INTERPRETATION OF A REGRESSION EQUATION

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. reg EARNINGS S
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Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
				Root MSE	=	13.126

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

This is the output from a regression of earnings on years of schooling, using Stata.

INTERPRETATION OF A REGRESSION EQUATION

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
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_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

For the time being, we will be concerned only with the estimates of the parameters. The variables in the regression are listed in the first column and the second column gives the estimates of their coefficients.

INTERPRETATION OF A REGRESSION EQUATION

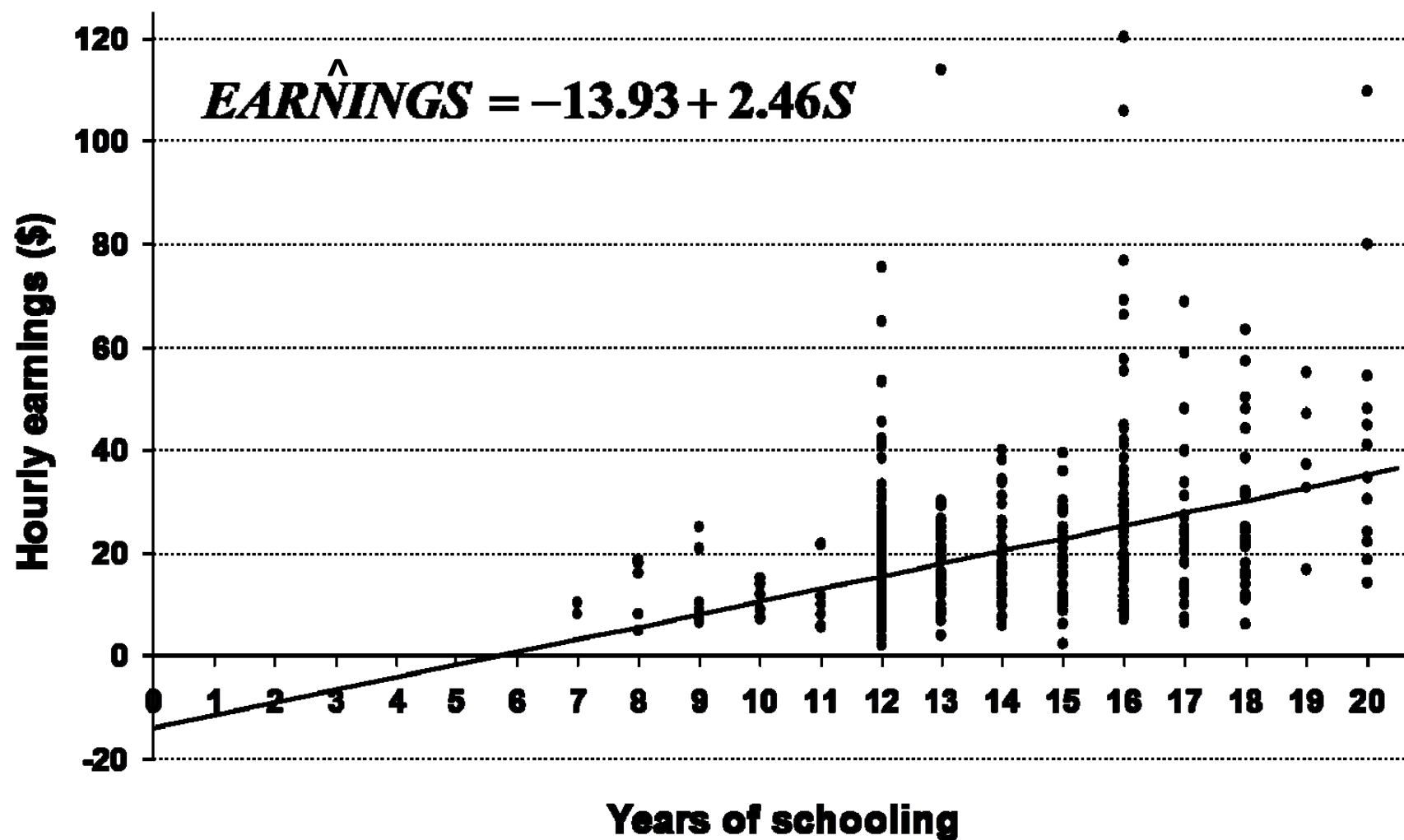
```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
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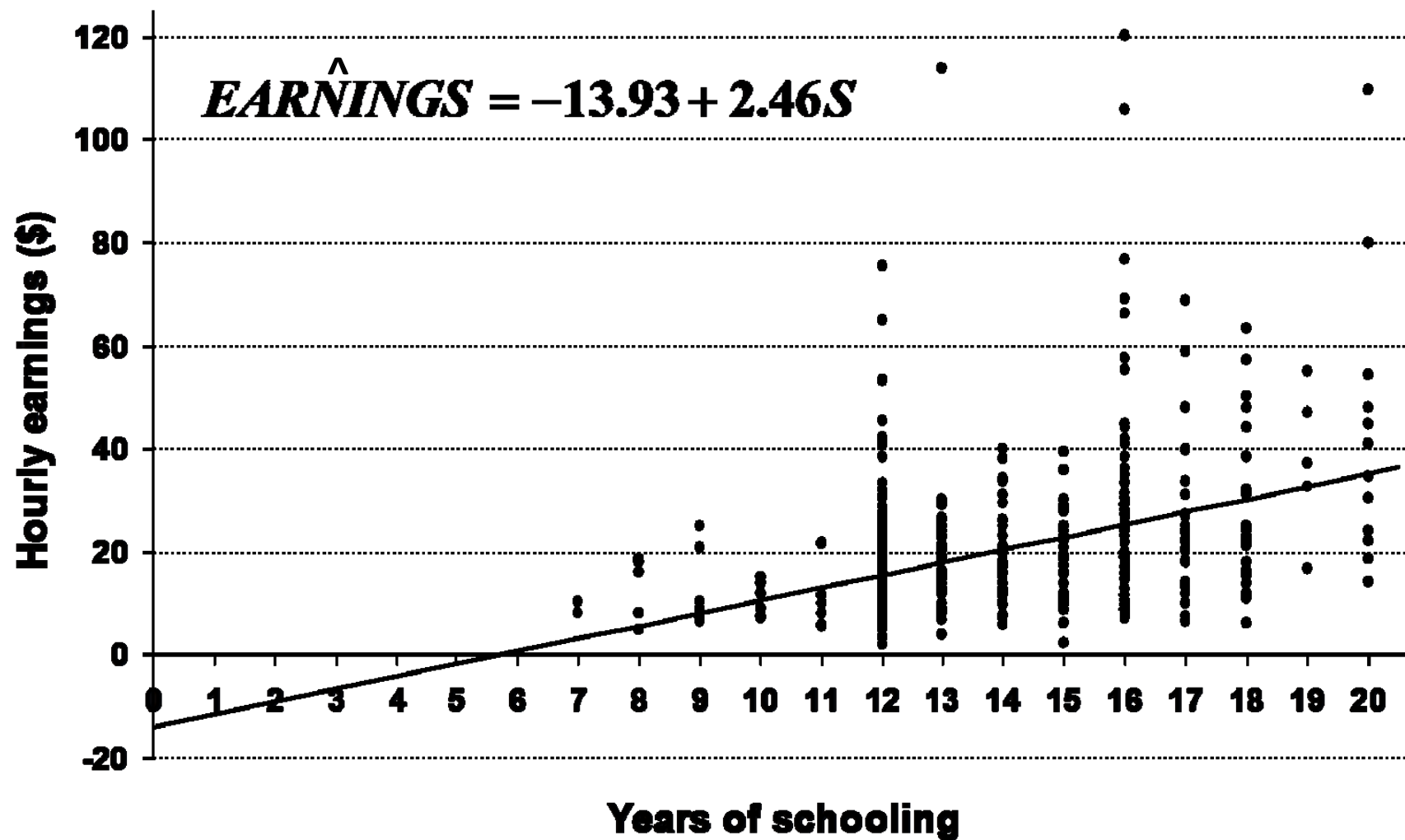
In this case there is only one variable, *S*, and its coefficient is 2.46. *_cons*, in Stata, refers to the constant. The estimate of the intercept is -13.93.

INTERPRETATION OF A REGRESSION EQUATION



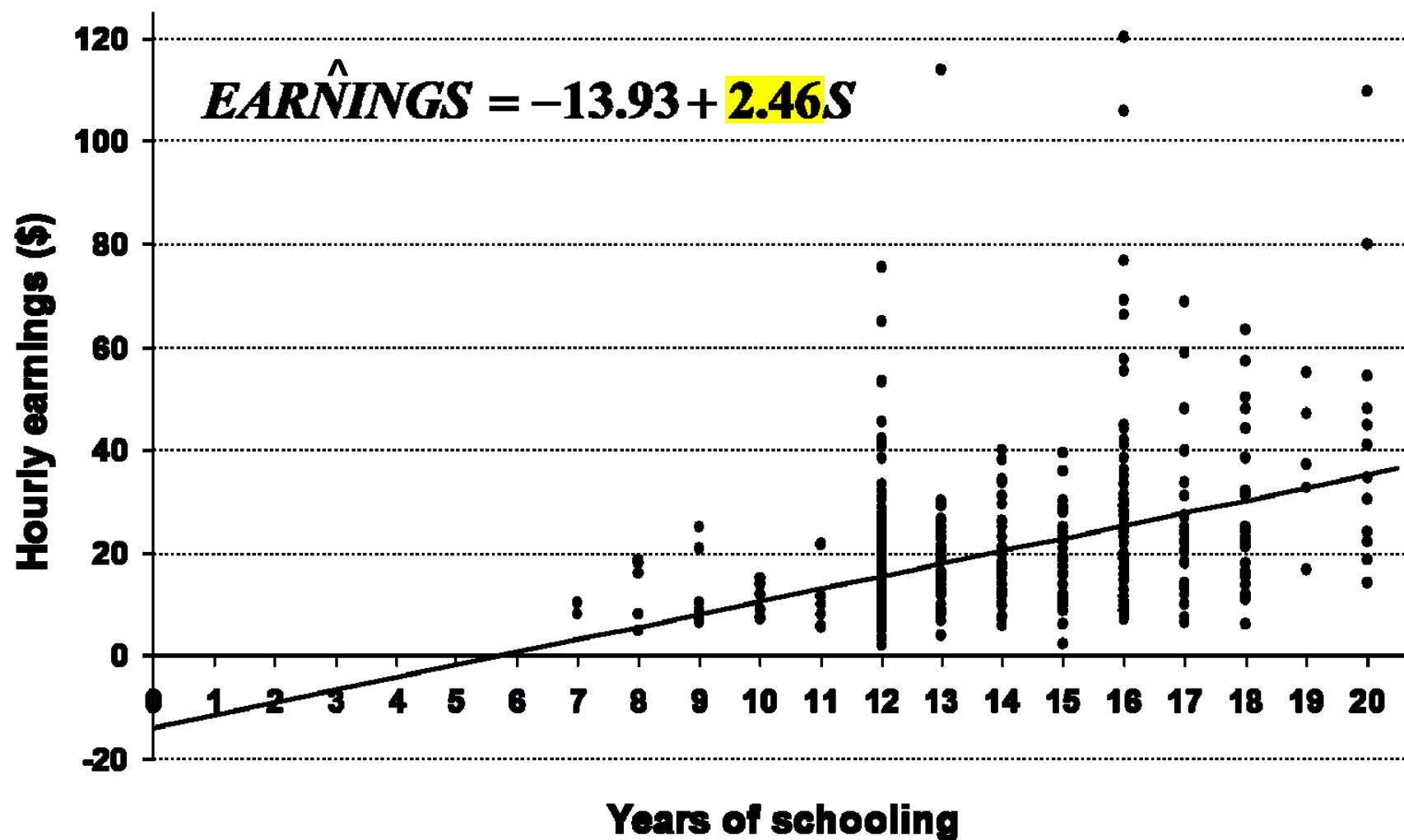
Here is the scatter diagram again, with the regression line shown.

INTERPRETATION OF A REGRESSION EQUATION



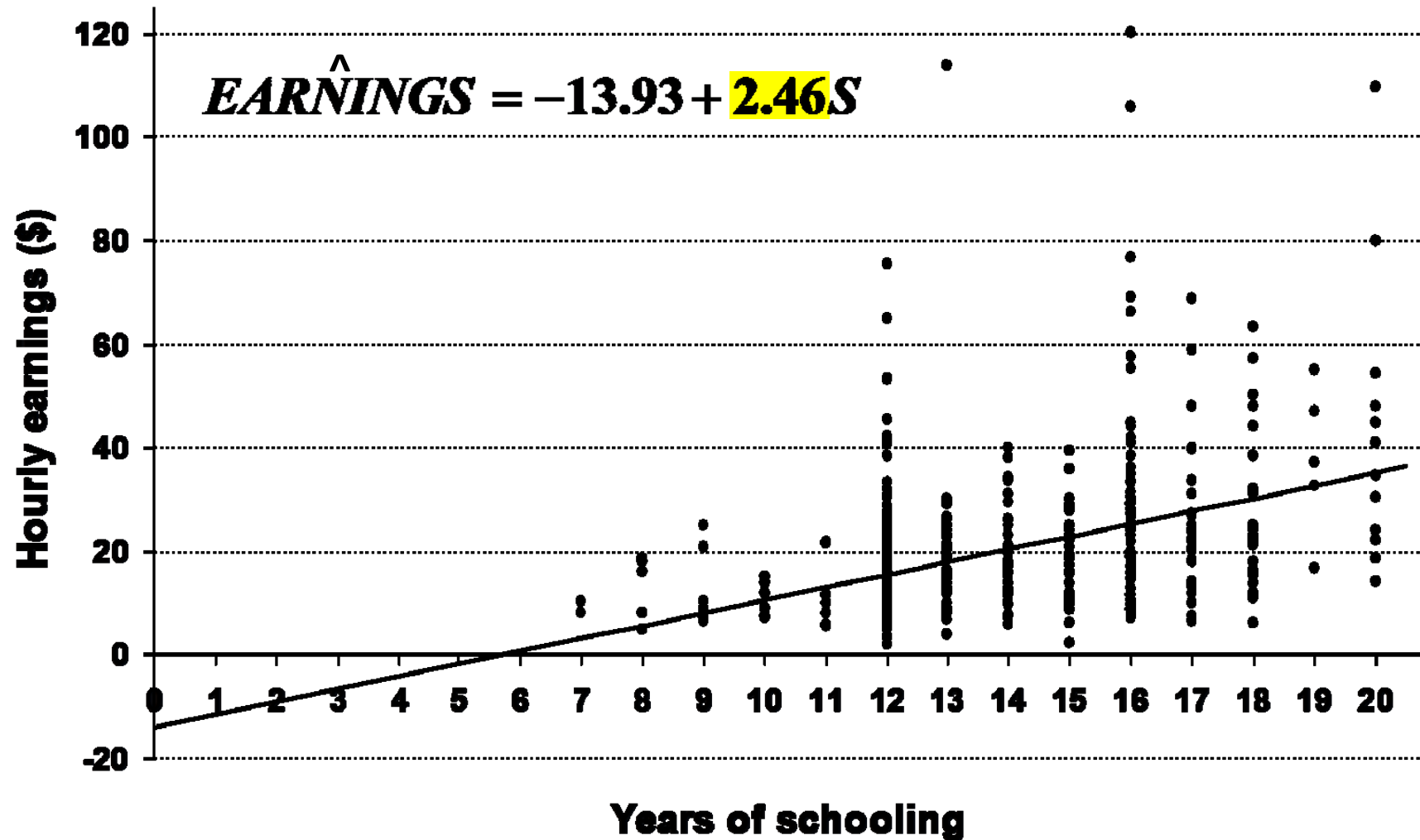
What do the coefficients actually mean?

INTERPRETATION OF A REGRESSION EQUATION



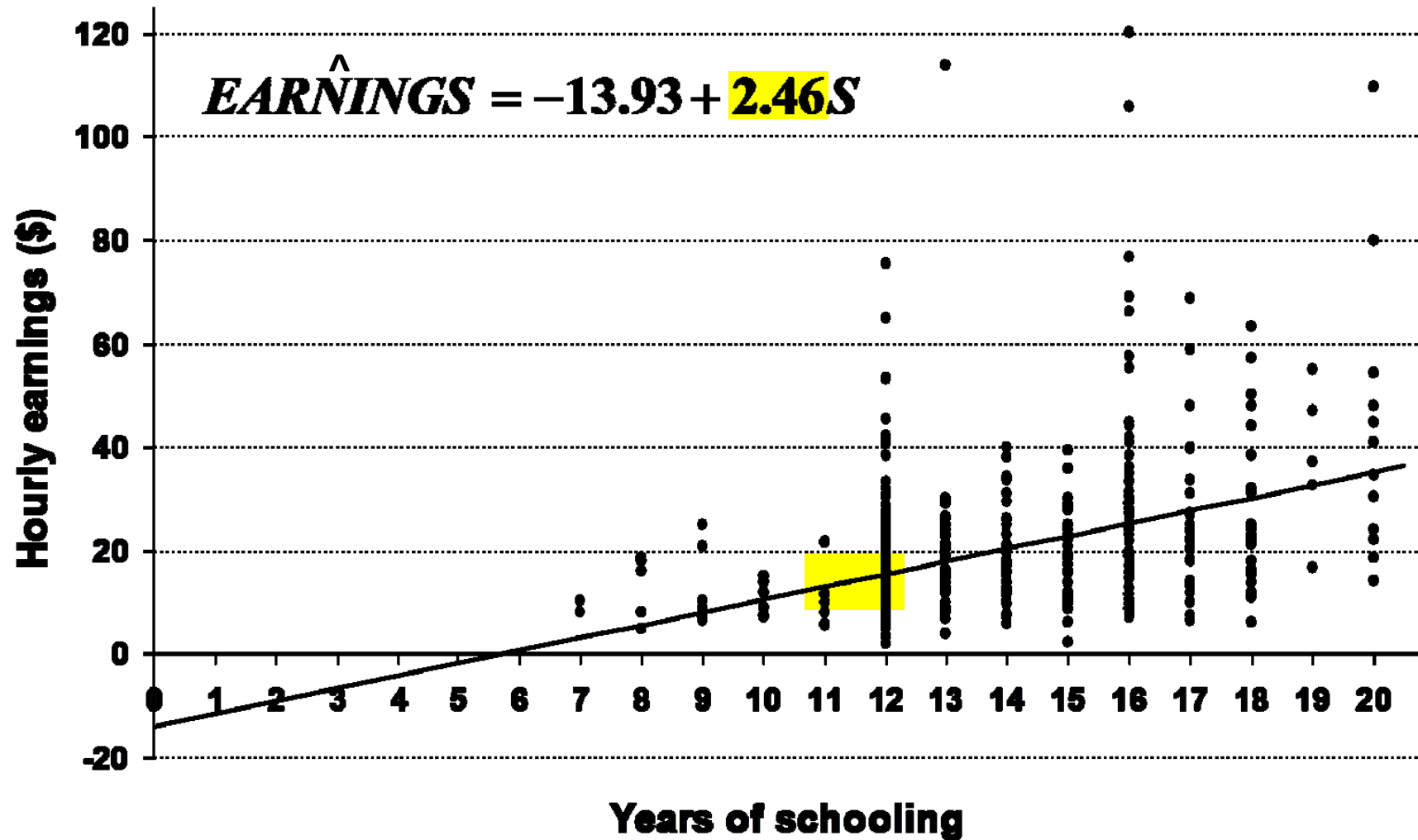
To answer this question, you must refer to the units in which the variables are measured.

INTERPRETATION OF A REGRESSION EQUATION



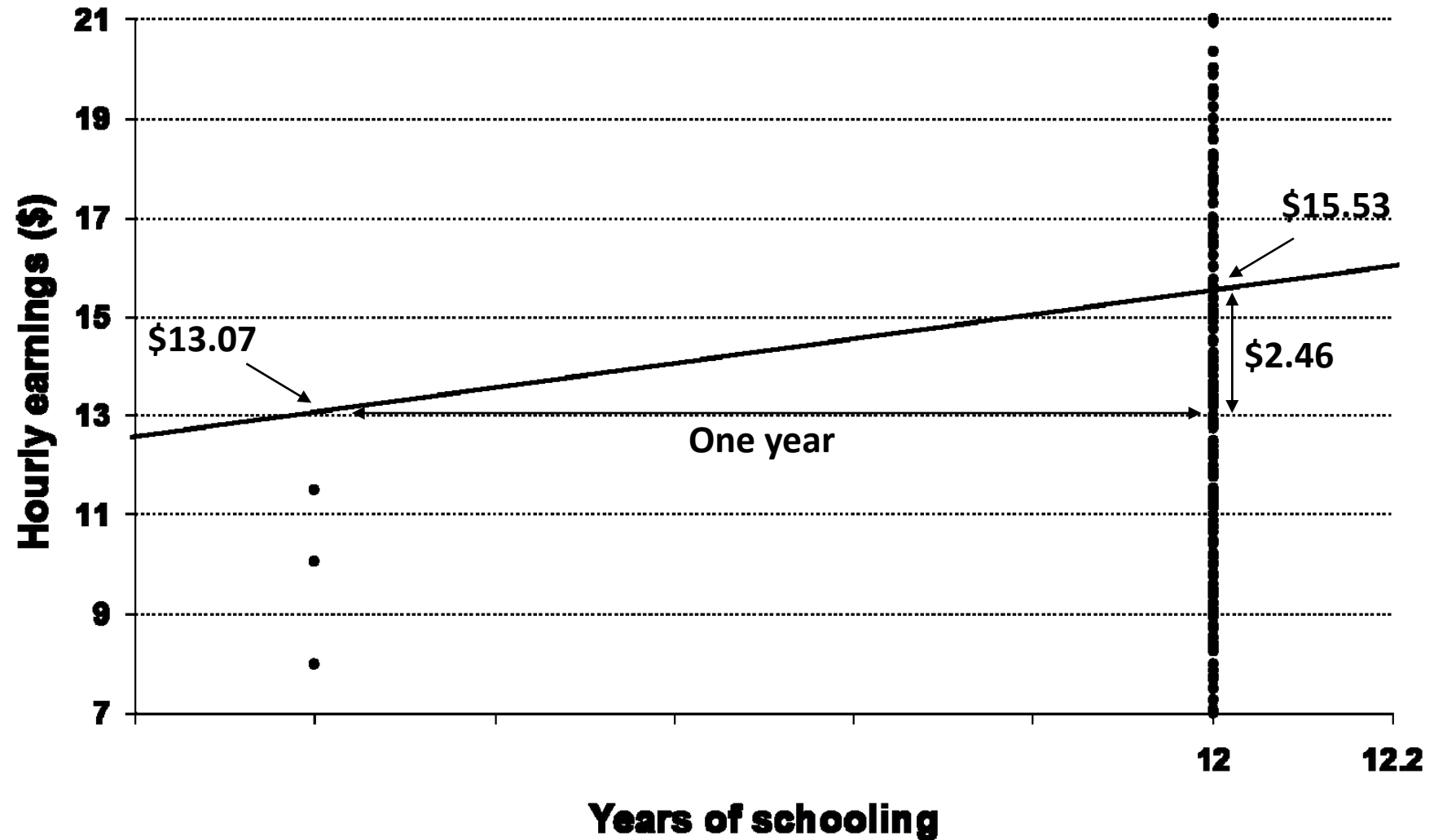
S is measured in years (strictly speaking, grades completed), $EARNINGS$ in dollars per hour. So the slope coefficient implies that hourly earnings increase by \$2.46 for each extra year of schooling.

INTERPRETATION OF A REGRESSION EQUATION



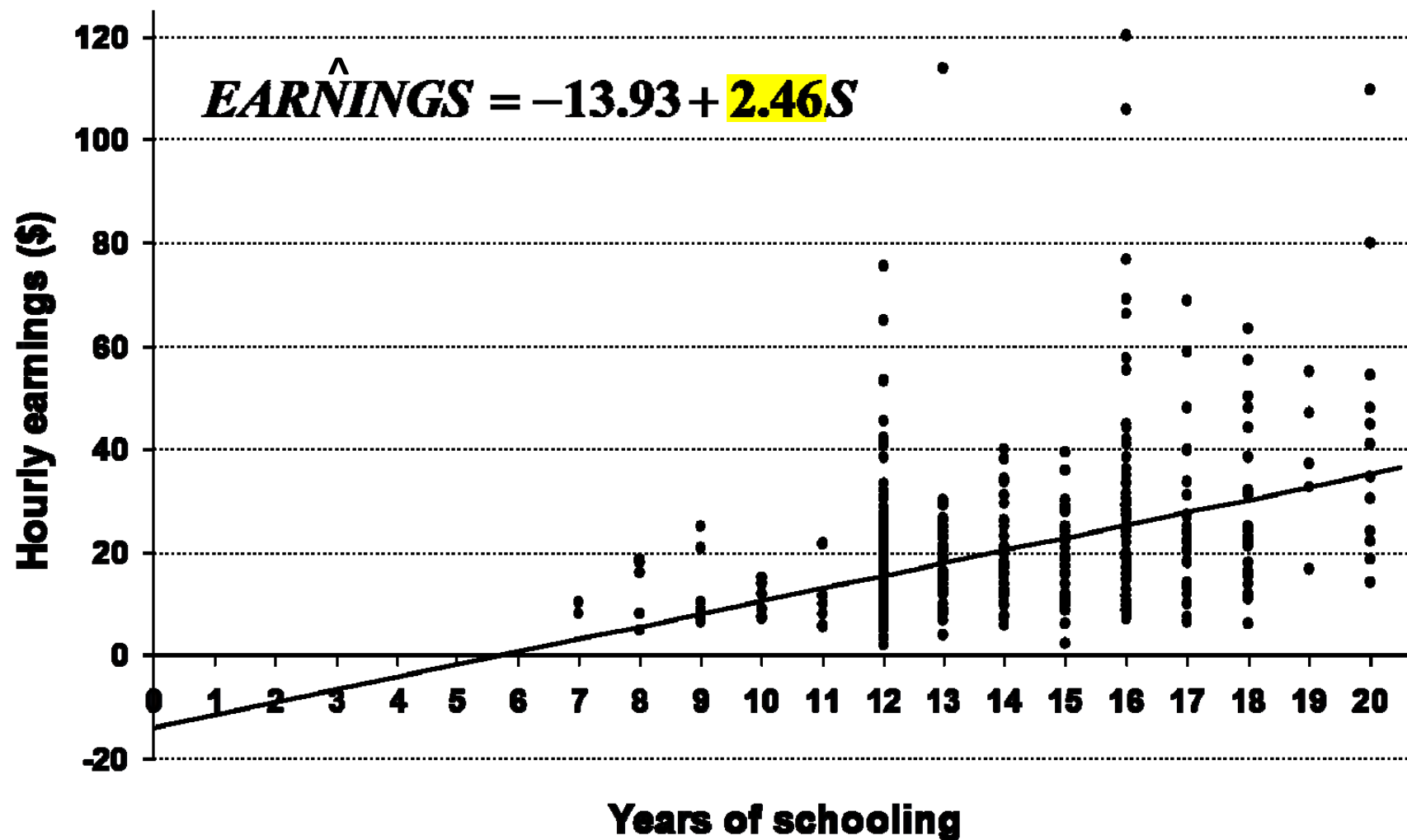
We will look at a geometrical representation of this interpretation. To do this, we will enlarge the marked section of the scatter diagram.

INTERPRETATION OF A REGRESSION EQUATION



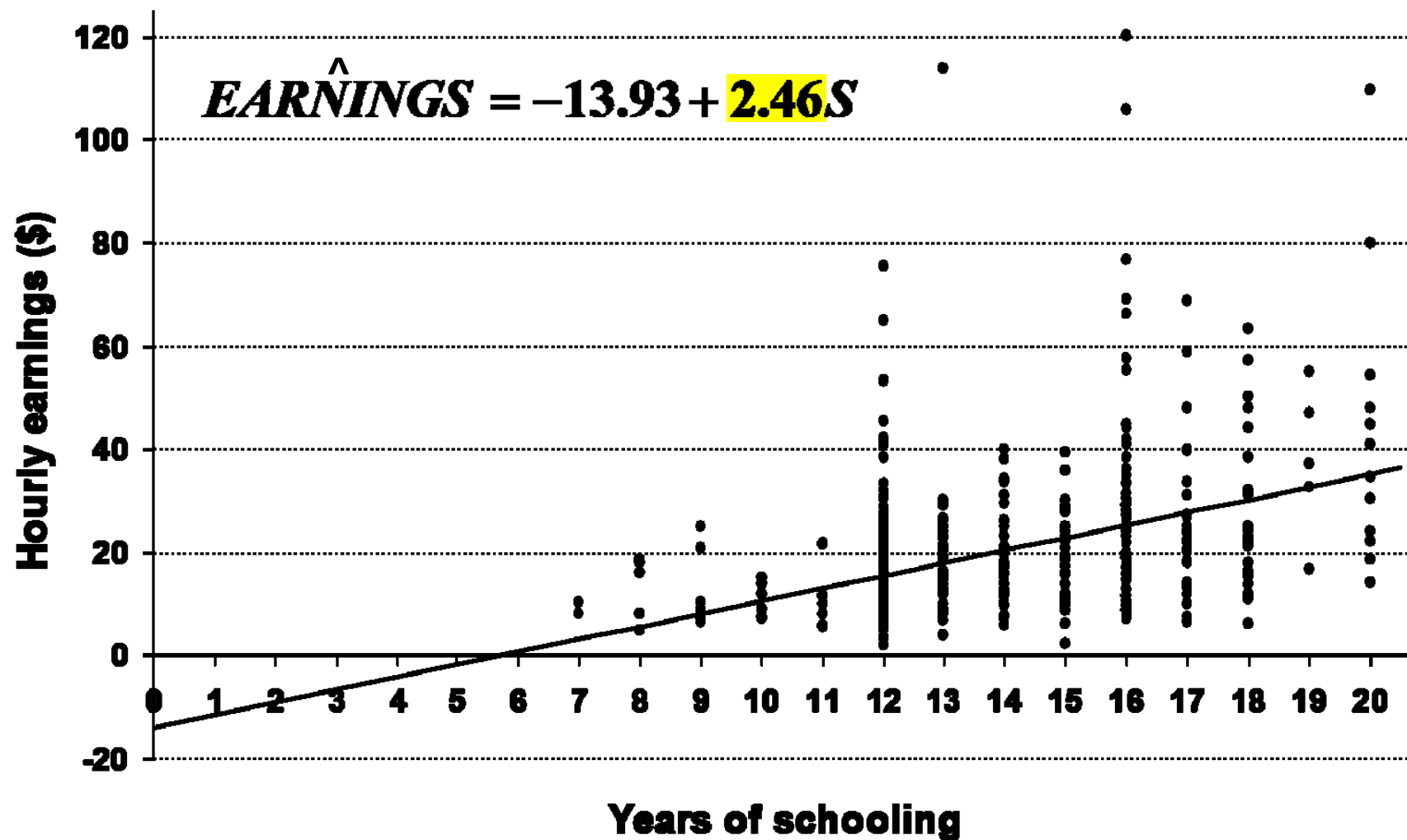
The regression line indicates that completing 12th grade instead of 11th grade would increase earnings by \$2.46, from \$13.07 to \$15.53, as a general tendency.

INTERPRETATION OF A REGRESSION EQUATION



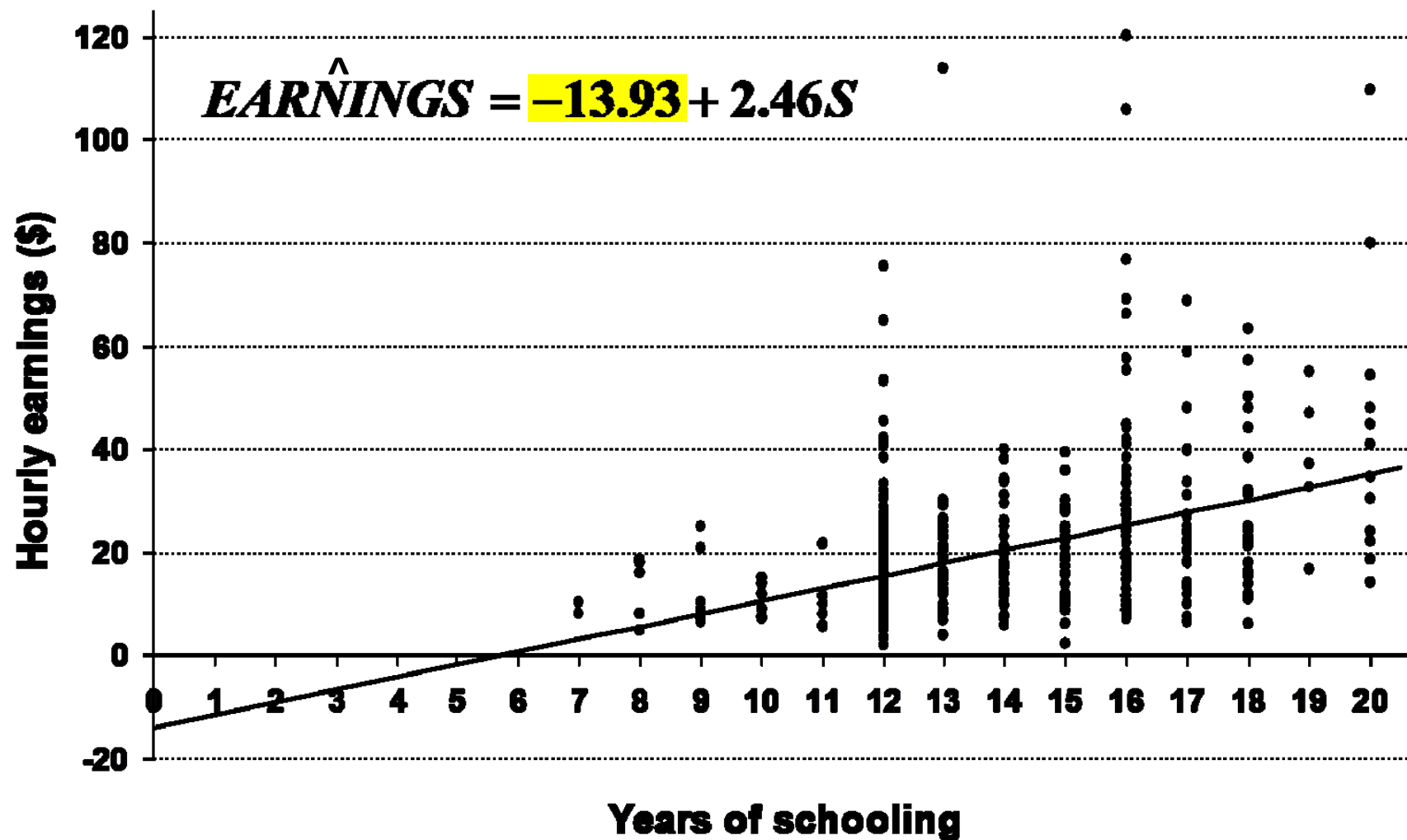
You should ask yourself whether this is a plausible figure. If it is implausible, this could be a sign that your model is misspecified in some way.

INTERPRETATION OF A REGRESSION EQUATION



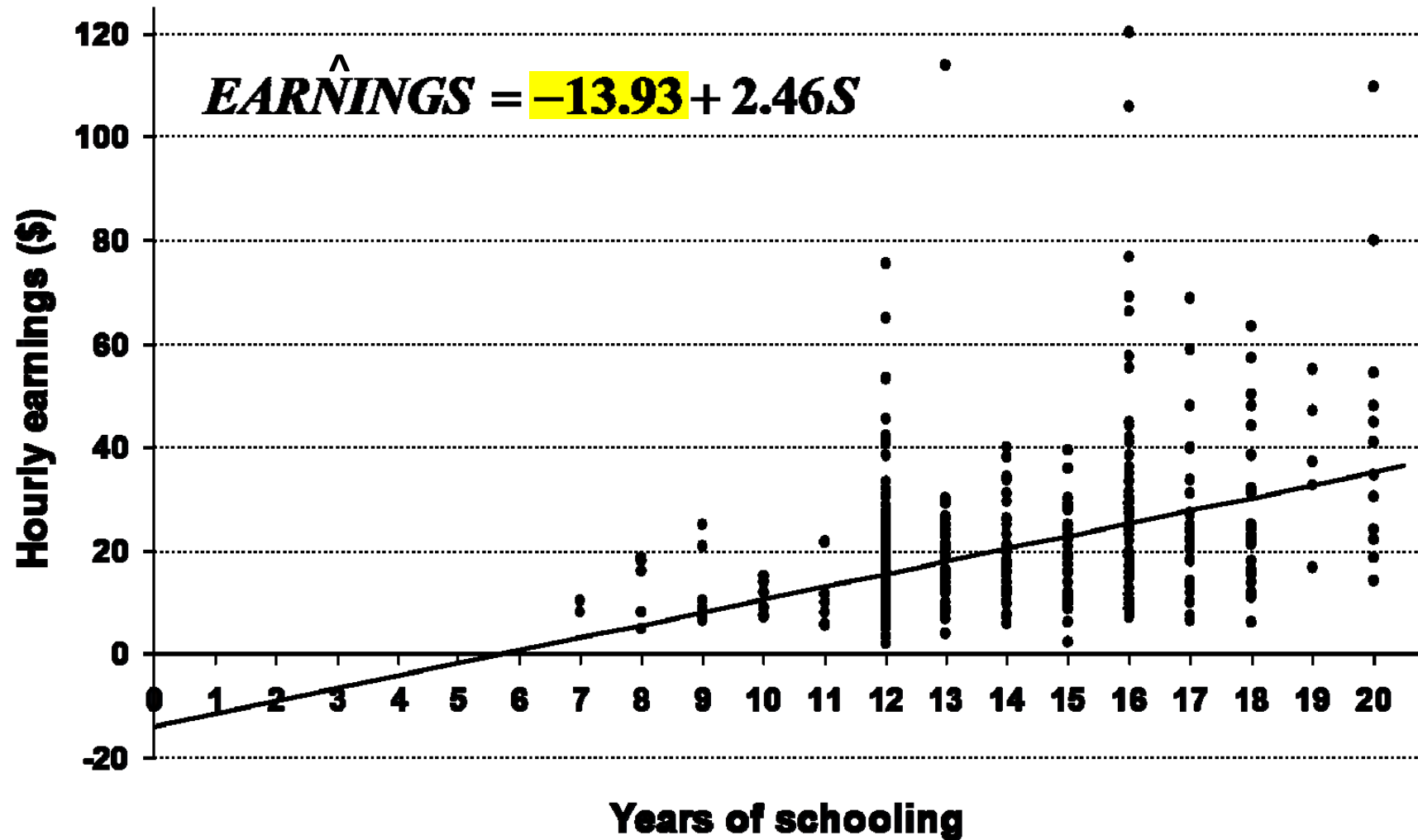
For low levels of education it might be plausible. But for high levels it would seem to be an underestimate.

INTERPRETATION OF A REGRESSION EQUATION



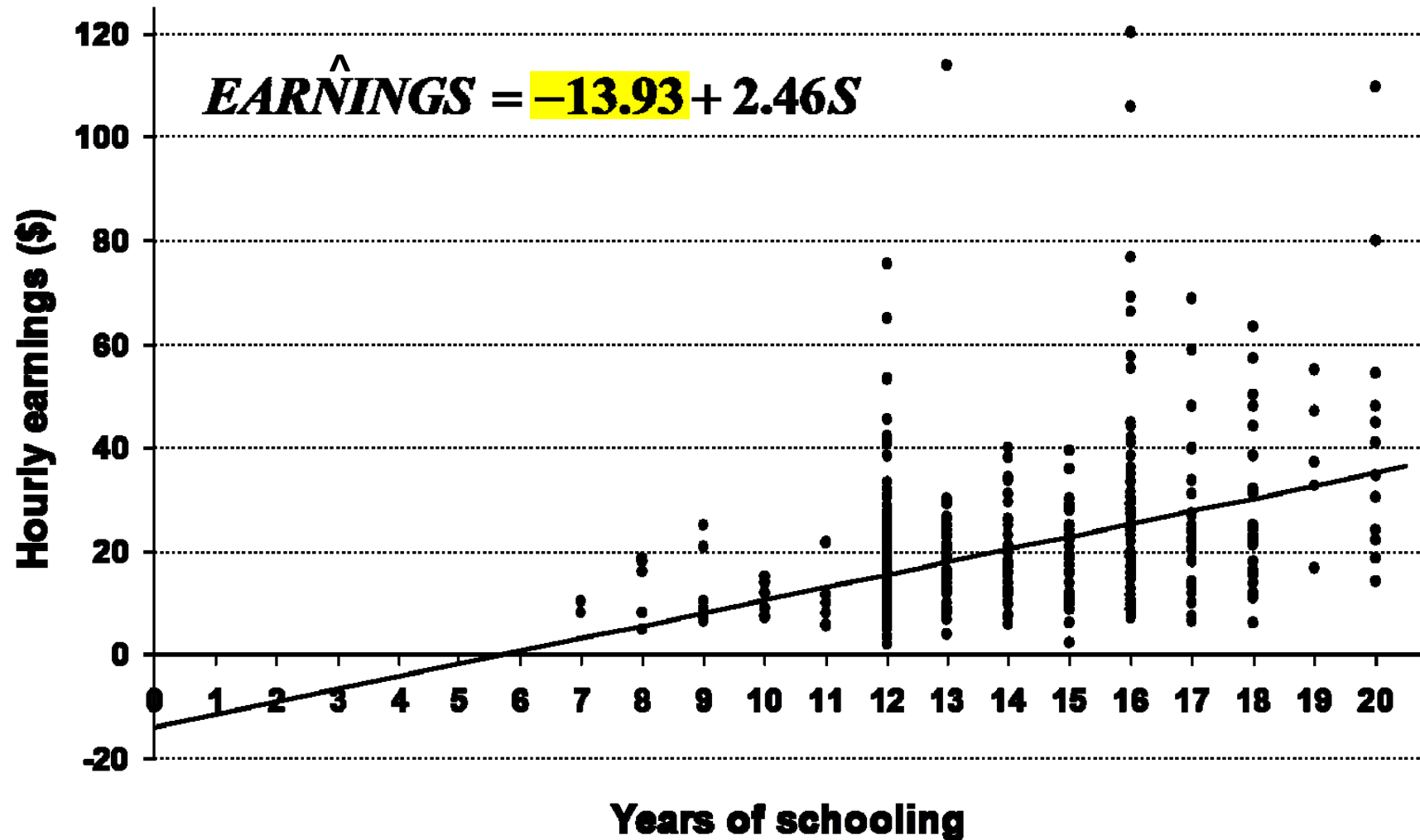
What about the constant term? (Try to answer this question yourself before continuing with this sequence.)

INTERPRETATION OF A REGRESSION EQUATION



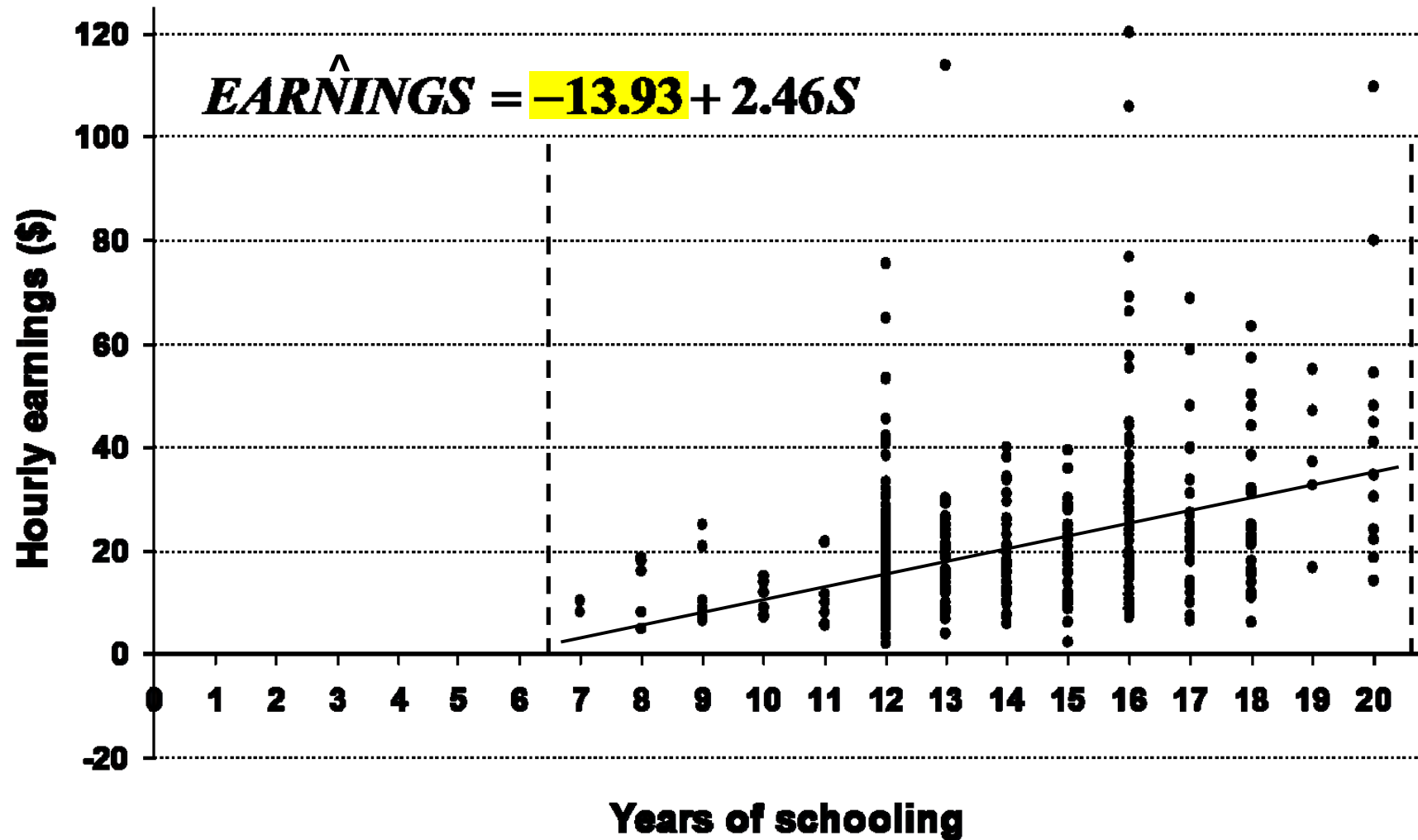
Literally, the constant indicates that an individual with no years of education would have to pay \$13.93 per hour to be allowed to work.

INTERPRETATION OF A REGRESSION EQUATION



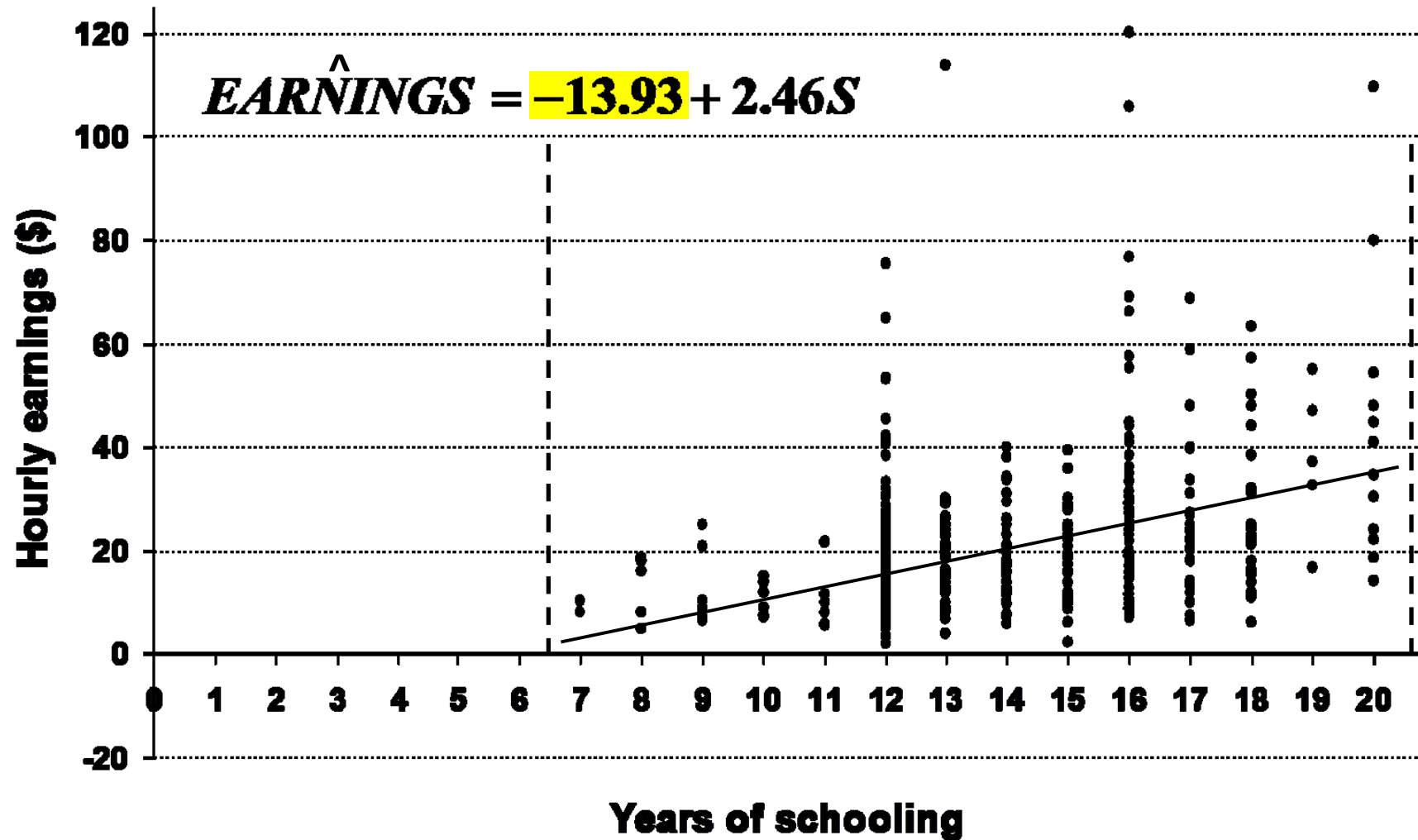
This does not make any sense at all. In former times craftsmen might require an initial payment when taking on an apprentice, and might pay the apprentice little or nothing for quite a while, but an interpretation of negative payment is impossible to sustain.

INTERPRETATION OF A REGRESSION EQUATION



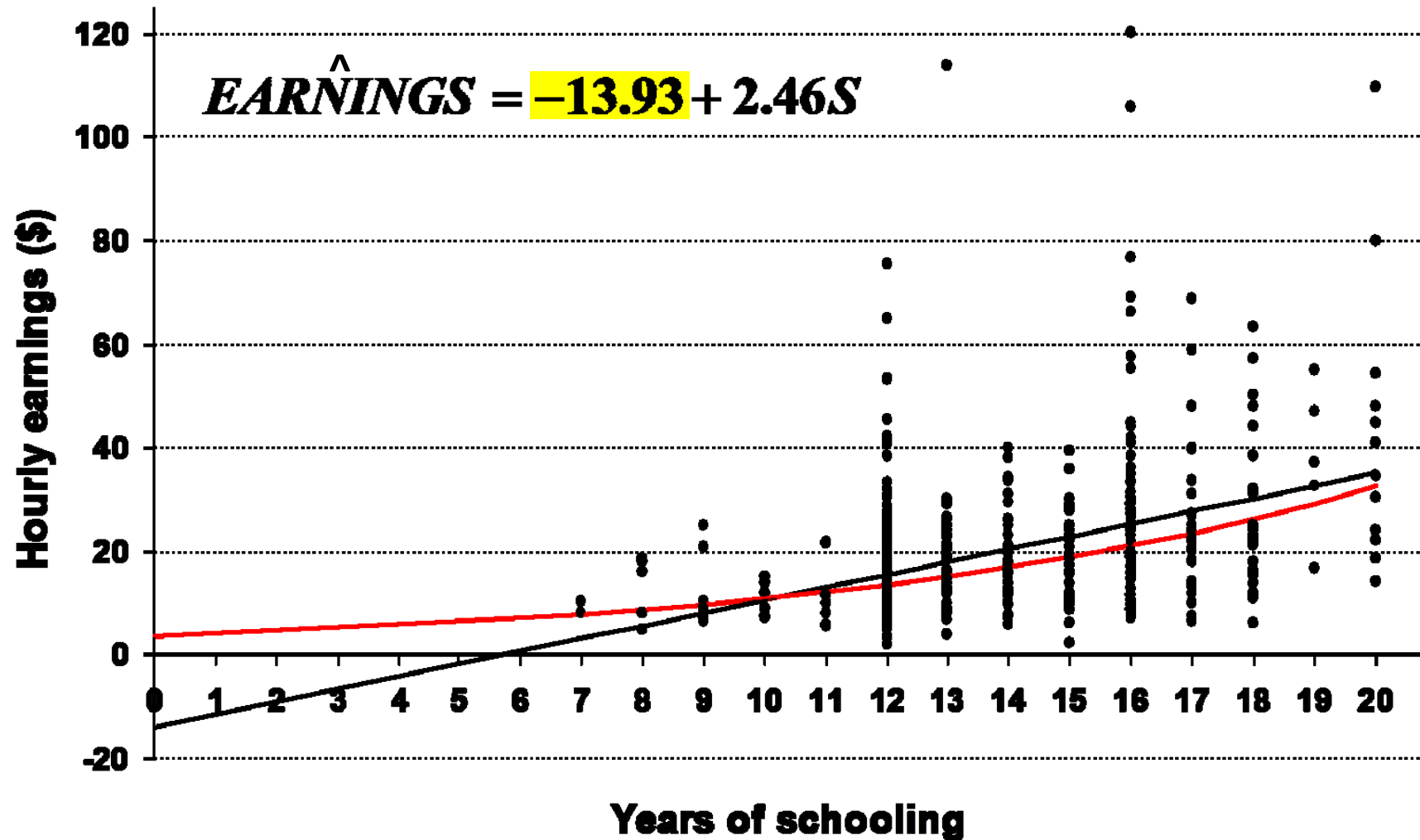
A safe solution to the problem is to limit the interpretation to the range of the sample data, and to refuse to extrapolate on the ground that we have no evidence outside the data range.

INTERPRETATION OF A REGRESSION EQUATION



With this explanation, the only function of the constant term is to enable you to draw the regression line at the correct height on the scatter diagram. It has no meaning of its own.

INTERPRETATION OF A REGRESSION EQUATION



Another solution is to explore the possibility that the true relationship is nonlinear and that we are approximating it with a linear regression. We will soon extend the regression technique to fit nonlinear models.